

Distortion of cortical hue geometry in color vision deficiency enables an individualized
color correction filter

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1 Abstract

Color vision deficiency (CVD) is typically corrected with filters that target a population-average retina and treat every user alike. Yet the cortical color representation that perception reads out is reshaped individually, and whether it could guide a personalized correction is unknown. Here we show, using fMRI in two adults with CVD studied as single cases alongside healthy controls, that the cortical representation of color is not reduced in overall signal but distorted in its response structure. Both CVD participants discriminated all displayed hues across the visual hierarchy, yet the structure of their color representation was distorted in two complementary ways. Continuous interpolation between hues was reduced at area hV4, significantly in the protan participant and as a same-direction trend in the deutan participant. Separately, the geometry of color representations was altered in early visual cortex (V1 and V2), differing between the two individuals. We then summarized each person’s distortion with a compact two-component cortical model and analytically inverted it into a per-individual stimulus-space filter, designed to move the CVD color perception and cortical representation structure toward that of a healthy observer. In a second session in the deutan participant, the personalized filter moved the impaired cortical hue interpolation partway toward the healthy-control reference — without reaching the control level — whereas a deployed accessibility filter did not improve it. This partial advantage was confined to the interpolation index: both filters restored behavioral discrimination comparably, and neither yielded an HC-like representational geometry. Quantitative fine-discrimination (2AFC) testing and replication in further individuals remain open. These findings recast CVD as an individually patterned distortion of cortical color structure rather than a loss of chromatic signal, one that can in principle be inverted into a personalized correction.

Keywords: color vision deficiency, anomalous trichromacy, fMRI, hV4, representational similarity analysis, forward encoding model, personalized color correction

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3 Introduction

Color vision deficiency (CVD) affects approximately 8% of men and 0.4% of women of European ancestry (Birch, 2012). It arises from altered spectral sensitivity of the L- or M-cone photopigment. This altered sensitivity is driven by X-linked polymorphisms in the cone opsin genes (Deeb, 2005; Neitz & Neitz, 2011). In anomalous trichromats, the deviant cone peak is displaced by roughly 2–12 nm from its normal position (Stockman & Sharpe, 2000). This small displacement reduces the L–M cone-opponent signal rather than abolishing it.

A weakened L–M signal yields a characteristic peripheral phenotype. Affected observers show reduced red–green discrimination, frequent category confusions, and elevated just-noticeable differences along the L–M chromatic axis. The Ishihara test screens for the resulting red–green confusions (Ishihara, 1917). The elevated discrimination thresholds are quantified instead by threshold psychophysics (Boehm, MacLeod & Bosten, 2014; Bosten, 2019).

Current CVD correction methods fall into two structurally related families. Hardware filters form the first family. Dichroic notch lenses, such as the EnChroma line, are designed to filter narrow spectral bands and amplify the residual red–green signal. Software filters form the second. Brettel–Viénot–Mollon simulation (Brettel, Viénot & Mollon, 1997) and the parameterized Machado retinal model (Machado, Oliveira & Fernandes, 2009) forward-project a stimulus onto an anomalous retina. Their Daltonization inverses, including recent algorithmic reformulations (Akalin et al., 2025), redistribute the lost contrast into channels the user can still see. Across this diversity, the two families share one commitment: they optimize against a population-average retina and apply the same transform to every user.

Attempts to escape this average-user ceiling have mostly taken one route: tuning a retinal cone-shift parameter to a user’s responses on plate-style or appearance-matching tests. The criterion is then the appearance *report*, not the cortical

representation that downstream tasks read out, so these methods personalize at the retinal rather than the cortical level. Their results show the cost. Improvements do not generalize, because the phenotype varies markedly even within one diagnostic category (Bosten, 2019). Color-enhancing glasses improved discrimination for only one of two consumer products tested (Patterson et al., 2022), and population-average filters shift the appearance of color while leaving generalized color discrimination largely unchanged (Somers, Franklin & Bosten, 2024). What is missing is a filter derived from, and grounded in, the user’s *own* neural color representation.

CVD is not an on–off loss of color. The downstream cortical color machinery stays intact. It receives an input that is quantitatively reweighted but qualitatively similar to a normal input. The cortical color representation is therefore not absent but structurally distorted. These features make the cortical representation the natural level at which to examine the distortion.

Cortical color is a distributed, hierarchical computation (Gegenfurtner, 2003; Shapley & Hawken, 2011; Conway, Eskew, Martin & Stockman, 2018). Early visual cortex carries cone-opponent signals. Higher-order regions, in particular hV4, support a population code for hue (Brouwer & Heeger, 2009), and act as a hub linking seen and remembered color (Bannert & Bartels, 2018). Continuous hue can be reconstructed from these distributed patterns (Brouwer & Heeger, 2009), and hue-selective responses to intermediate colors appear as early as V1 (Parkes, Marsman, Oxley, Goulermas & Wuerger, 2009; Kuriki, Sun, Ueno, Tanaka & Cheng, 2015). This code is shared enough across people that one observer’s color responses can be read out in a common space, an approach recently validated in healthy observers (Bannert & Bartels, 2025).

Anomalous trichromats do more than lose sensitivity. They also compensate. Long-term cortical adaptation reshapes hue appearance independently of the retina (Boehm et al., 2014; Webster, 2015; Tregillus et al., 2021). In deutan observers, chromatic-contrast adaptation exceeds what threshold sensitivity loss predicts, consistent with a post-receptoral gain that partly offsets the weaker L–M signal (Basim, Goddard, Yang & Webster, 2025). Chromatic responses also realign toward the

blue–yellow direction in these observers (Emery, Volbrecht, Peterzell & Webster, 2021). Because this evidence comes from a separate paradigm, it motivates rather than establishes a cortical locus, and we do not replicate it here. Together these findings locate key variance at the retinal–cortical interface, not at the retina alone.

Because this compensation lives in cortex, a correction calibrated to the retina or to appearance reports inherits it and acts at the wrong level. We therefore take the individual’s cortical readout, not the simulated retina, as the reference for correction. This is a design hypothesis, and the rest of the paper tests it.

Three concrete gaps remain. (*Gap 1*) Neural studies of color describe geometry at the group level in healthy observers (Brouwer & Heeger, 2009; Kuriki et al., 2015; Bannert & Bartels, 2018). The cortical color geometry of an individual CVD observer, the personal trait on which a tailored correction would be built, has not been characterized. (*Gap 2*) The measures applied to CVD do not capture this geometry. fMRI in CVD has quantified response magnitude: contrast-response gain in anomalous trichromacy (Tregillus et al., 2021), and activation in dichromacy and achromatopsia (Rina et al., 2024). Where multivoxel patterns have been analyzed, the readout has been classification accuracy. Magnitude indexes signal strength and accuracy indexes category identity, but neither preserves the continuous relational structure (Brouwer & Heeger, 2013) that a hue-space filter must reshape. (*Gap 3*) Across studies that analyze multivoxel color patterns, we are not aware of a corrective filter inverted from an individual’s own cortical representation. Whether such a high-dimensional signal can be turned into a realizable, individualized correction is untested.

We ask whether an individual’s cortical color geometry can serve as the reference for a personalized correction. To answer this, we characterize two adults with CVD, one with each major red–green subtype, against a reference group of healthy controls (see Methods). Observations from the single protan participant are exploratory and need independent replication.

A stimulus-space filter is both warranted and tractable only in a specific regime.

The displayed colors must stay neurally distinguishable, or the filter has no signal to act on. The continuous arrangement of hues must also be distorted, or there is nothing to correct. Our questions follow this logic.

1. **Describe.** Does an individual’s cortical hue geometry in CVD carry structured distortion, with categorical discrimination preserved but the continuous arrangement of hues impaired?
2. **Summarize.** Can that distortion be captured by a few interpretable parameters, treated as a compact description rather than as physiological measurements?
3. **Correct.** Can a stimulus-space correction be derived from those parameters as a realizable, individualized filter?
4. **Validate.** Does the filter move the cortical representation toward the healthy-control reference, more so than a deployed accessibility filter?

We expect the continuous arrangement of hues to be selectively impaired while categorical discrimination is preserved, with the largest distortion among the blue-to-magenta hues. A few interpretable parameters should capture this distortion, and inverting them should yield a realizable, individualized correction.

Together, these steps yield a filter built for one individual. Its input is the cortical signal, its parameters are the fitted description, and its output is a stimulus shift. Each is defined at the level of the single observer, the level at which any correction must ultimately act. Because the sample holds one participant per subtype, this study tests the feasibility of the inversion: whether an individual’s cortical color geometry can be turned into a realizable filter. It does not test whether per-person correction outperforms a subtype average, which would require several individuals within one subtype.

4 Methods

The analysis follows three stages. First, two neural measures characterized each CVD participant’s hue-processing deficit: leave-one-color-out (LOCO) decoding at hV4

yielded a per-hue vulnerability profile (§4.7), and the between-group RDM difference (ΔRDM) captured the per-pair geometric deviation of CVD representational geometry from the HC mean (§4.8). Second, two candidate CVD distortion model classes were fit by minimizing a combination of neural and behavioural losses, with parameters selected by a pre-specified stability procedure (§4.9–§4.11). Third, the fitted distortion was analytically inverted to yield a per-subject stimulus-space correction filter (§4.13); Figure 3 summarizes this fit-and-invert workflow. This filter was then evaluated in a repeat fMRI session and a behavioural discrimination battery (JND thresholds and 8AFC color identification; §4.14).

4.1 Participants

Thirteen volunteers were recruited under Institutional Review Board approval (#2510/002-023) and provided written informed consent. Color vision was screened using the 14-plate Ishihara pseudoisochromatic test (Ishihara, 1917). Participants identifying ≥ 9 plates correctly were classified as healthy controls (HC; $n = 7$, 3 female, age 22.7 ± 2.5 years). Two male participants scoring below this threshold were classified as CVD: the deutan participant (deuteranomalous, 5/14 plates corrected) and the protan participant (protanomalous, 7/14 plates corrected). Four additional participants were excluded for failing to complete the full scanning protocol. All CVD results are interpreted as single-case demonstrations using the Crawford and Howell (1998) modified t -test; no population-level inference is made.

4.2 Stimuli and task

Eight isoluminant chromatic stimuli were drawn from CIE $L^*a^*b^*$ space at equal angular spacing (0° – 315° , 45° steps; $L^* = 75$, chroma = 40) (Commission Internationale de L’Eclairage, 1986), spanning red, orange, yellow, green, cyan, blue, purple, and magenta (Figure 1A). A gray filler ($a^* = b^* = 0$) served as a null condition.

Stimuli were presented in a Rapid Serial Visual Presentation (RSVP) design to sustain attention and fixation. Each stimulus was a uniform circular chromatic patch (a colored disc, 1.5 s per stimulus) shown in pseudo-random order optimized with

Neurodesign (Durnez, Moerkerke & Nichols, 2018). Participants pressed a button upon detecting two consecutive 'K's within a rapid serial letter stream presented at the centre of the chromatic disc. Each run contained eight repetitions of all eight chromatic stimuli and the gray filler; inter-stimulus intervals were drawn from $\{3, 4.5, 6\}$ s. Each session comprised six runs (≈ 7 min each). The experiment was coded in PsychoPy 2022.2.5 (Peirce et al., 2019).

All participants completed JND behavioral testing in the same scanning session as the main MRI acquisition. The deutan participant additionally completed a second session in which JND was repeated under the per-subject stimulus-space filter, alongside a second fMRI acquisition (see the Behavioral tasks section).

4.3 MRI acquisition and preprocessing

Data were acquired on a Siemens 3T MAGNETOM Cima.X scanner with BioMatrix 3T coils. To constrain coverage to posterior occipital cortex we used 24 oblique slices perpendicular to the calcarine sulcus (Brouwer & Heeger, 2009) (TR 1.5 s, TE 30 ms, FA 75° , $2 \times 2 \times 2$ mm, 96×80 matrix).

Data were converted to BIDS format and defaced with ezBIDS (Levitas et al., 2024). Functional images were registered to the T1w image via mutual-information coregistration (FreeSurfer `mri_coreg`; Wells, Viola, Atsumi, Nakajima & Kikinis, 1996; Maes, Collignon, Vandermeulen, Marchal & Suetens, 1997). T1w-to-MNI normalization used 12-DOF affine (FSL FLIRT; Jenkinson, Bannister, Brady & Smith, 2002) followed by nonlinear warping (FNIRT; Andersson, Jenkinson & Smith, 2007), yielding 2 mm isotropic MNI-space BOLD data.

4.4 ROI definition and response estimation

Bilateral V1, V2, V3, and hV4 were defined from the Wang probabilistic atlas (Wang, Mruczek, Arcaro & Kastner, 2015) at $> 50\%$ probability, intersected with each participant's BOLD brain mask (voxel counts: V1 655 ± 214 ; V2 451 ± 145 ; V3 103 ± 29 ; hV4 63 ± 22 ; mean $\pm$ SD).

Single-trial response amplitudes were estimated with a two-stage GLM (Dale, 1999; Brouwer & Heeger, 2009, 2013). First, an ROI-level hemodynamic response function (HRF) was extracted from a finite-impulse-response (FIR) model (8-TR basis set). Voxels in the top 50% of variance explained by the mean ROI HRF were retained (V1 328 ± 107 ; V2 226 ± 73 ; V3 52 ± 15 ; hV4 32 ± 11 voxels). Second, the estimated HRF was convolved with per-color design matrices and fitted by ridge regression to yield one amplitude per (voxel, color, run), organized as a $6 \times 8 \times V$ array per subject.

Voxel amplitudes were aligned across runs using within-subject Procrustes rotation (Gower, 1975): runs 2–6 were mapped to run 1 via the orthogonal transformation minimizing $\|X_{\text{ref}} - XQ\|_F$ (scaling prohibited).

4.5 Shared Response Model

Cross-subject alignment and dimensionality reduction were performed with the Shared Response Model (SRM; Chen et al., 2015). SRM estimates per-participant orthonormal mappings $W_i \in \mathbb{R}^{V \times k}$ (V voxels, k latent dimensions) and a group-shared response matrix $S \in \mathbb{R}^{k \times 8}$:

$$\min_{W_i, S} \sum_{i=1}^N \|X_i - W_i S\|_F^2 \quad \text{s.t.} \quad W_i^\top W_i = I_k \quad (1)$$

SRM was trained exclusively on the seven HC participants to avoid circular inference in CVD comparisons. Each CVD participant was subsequently projected into the fixed HC-derived space by solving $W_{\text{CVD}} = UV^\top$ where $U\Sigma V^\top = \text{SVD}(X_{\text{CVD}} \cdot S^\dagger)$. Reduced dimensions were chosen by mean-rank aggregation across seven-fold leave-one-subject-out (LOSO) cross-validation, using three SRM-quality metrics (RDM split-half reliability, cross-subject RDM correlation, and reconstruction error): V1 $k = 4$; V2 $k = 4$; V3 $k = 3$; hV4 $k = 3$.

4.6 Forward encoding model

Both discrimination (LORO) and interpolation (LOCO) decoding used the same forward encoding model (Brouwer & Heeger, 2009). The model represents each stimulus

as a $F = 6$ -dimensional channel vector $\mathbf{c}(\theta) = [c_1(\theta), \dots, c_F(\theta)]^\top$, where $c_k(\theta) = \max(0, \cos(\theta - \theta_k))^2$ is a half-wave rectified squared cosine tuned to preferred hue θ_k (preferred hues spaced equally at 60° ; Brouwer & Heeger, 2009). Decoding and encoding optimize different targets and therefore use different estimators of the channel-to-voxel weight matrix $W \in \mathbb{R}^{F \times V}$.

Decoding (stimulus hue from voxels). Decoding recovers the stimulus hue from voxel responses. We estimated W by ordinary least squares (pseudoinverse) and inverted it. For a held-out voxel pattern $\mathbf{x}$, the channel activations were reconstructed as $\hat{\mathbf{c}} = W\mathbf{x}$, and the returned hue was the one whose basis vector had the highest Pearson correlation with $\hat{\mathbf{c}}$ (Brouwer & Heeger, 2009; Figure 2). Alternative decoders evaluated during model selection are described in Appendix A.

Encoding (voxel responses from channels). Encoding predicts the voxel responses from the channels ($\hat{X} = CW^\top$). We estimated W by ridge regression, with the penalty α selected per voxel by generalized cross-validation (GCV; Golub, Heath & Wahba, 1979; see Supplementary §S6). Regularization is warranted because the few stimuli and the correlated basis channels would otherwise overfit W and degrade prediction of held-out responses (Kay, Naselaris, Prenger & Gallant, 2008; Naselaris, Kay, Nishimoto & Gallant, 2011). This ridge prior is isotropic and imposes no spatial structure on the voxels.

4.7 Two decoding schemes: cross-run discrimination and cross-color interpolation

Two complementary cross-validation schemes used the forward encoding model to characterize each participant’s hue processing, differing only in what was held out. Leave-one-run-out (LORO) held out repeated presentations to test categorical discrimination, and leave-one-color-out (LOCO) held out entire hues to test continuous interpolation.

Cross-run discrimination (LORO). Categorical discrimination was assessed with leave-one-run-out cross-validation in SRM-aligned space. For each of the six runs, W was estimated on the remaining five runs using mean-subtracted SRM

projections, and each held-out run’s 8 color patterns were decoded by nearest-neighbour matching in channel-output space (the highest-correlation readout of §4.6). Primary discrimination performance was the 8-class exact accuracy (chance = $1/8 = 0.125$), averaged across six folds.

Group-level HC–CVD comparison used a Mann–Whitney U test on cross-subject generalization scores: the encoding model trained on each HC participant’s SRM-aligned data was applied to every other participant’s aligned patterns, yielding 21 unique HC-to-HC and 14 HC-to-CVD pairwise generalization scores. Within-ROI single-case tests used the Crawford and Howell (1998) modified t -test (one-tailed, lower).

Cross-color interpolation (LOCO). Continuous hue interpolation was assessed with leave-one-color-out cross-validation. Each of the 8 hues was held out in turn; W was estimated on the remaining 7 hues using all 6 runs (pooled 42-sample design matrix), and the held-out hue was predicted by nearest-neighbour matching in channel-output space.

Adjacent accuracy. The primary cross-color decoding metric was adjacent accuracy: the proportion of predictions that fell within ± 1 hue step (45°) of the target hue, averaged across the 6 runs of each test color (range 0–1; chance level under a uniform random predictor = $3/8 = 0.375$; exact-accuracy chance = $1/8 = 0.125$). For HC above-chance testing and for pairwise HC–CVD comparisons at hV4, p -values were obtained from 1,000 random color-label permutations and from the Crawford and Howell (1998) modified t -test (one-tailed, lower), respectively.

Vulnerability profile. The 8-dimensional vector of per-hue adjacent-accuracy values, $\mathbf{v} \in [0, 1]^8$, constitutes each participant’s *hue vulnerability profile*. For group-level model fit, we report Spearman ρ between the observed $\mathbf{v}$ and the $\mathbf{v}$ predicted by the CVD distortion model (see the Candidate CVD distortion models section). Per-hue group comparisons (Figure 4C) used the Crawford and Howell (1998) modified t -test (one-tailed, uncorrected). Effect sizes are reported as Hedges’ d (Hedges & Olkin, 1985) with the Crawford–Howell correction for small- n comparisons.

4.8 Between-group representational dissimilarity

To characterize the representational geometry of LOCO impairment in SRM-aligned space, we computed two measures. Mean pairwise disparity captures whether CVD geometry is globally more dispersed than HC. The between-group RDM difference, ΔRDM , captures which color pairs drive the deviation and tests whether a cortical model of the CVD distortion accounts for the pattern.

Pairwise disparity. For each participant and ROI, the RDM was the 8×8 matrix of pairwise correlation distances ($1 - \text{Pearson } r$). Mean pairwise disparity was the average of the 28 upper-triangle elements. To provide an unbiased HC reference for the held-out HC participant, the HC disparity distribution was estimated by a leave-one-subject-out (LOO) procedure. Under the directional prediction that CVD geometry is more dispersed than HC, individual CVD disparity was compared to the HC LOO distribution by the Crawford and Howell (1998) modified t -test (one-tailed, upper; $\alpha = 0.05$). In this primary estimate the shared SRM space is trained on all seven HC, so each HC reference is in-sample whereas the projected CVD participant is out-of-sample. To control for this asymmetry we additionally computed a symmetric leave-one-subject-out (LOSO) estimate in which the shared space is retrained on six HC and the held-out HC is projected by SVD identically to each CVD participant, with the same Crawford–Howell test applied. The all-HC estimate is primary because every other analysis in the pipeline (LORO, LOCO, ΔRDM , and the inverse fit) operates in that same space; the symmetric LOSO is reported as a disclosed sensitivity analysis (table S1).

Geometric deviation from HC and model correspondence. The per-pair deviation of each CVD participant’s RDM from the HC mean was $\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC}}$, where $\overline{\text{RDM}}_{\text{HC}}$ is the grand mean RDM over the seven HC participants. Correspondence between the observed ΔRDM and the prediction of the two-component CVD distortion model (see Candidate CVD distortion models) was reported descriptively as the Spearman ρ between the upper-triangle elements of the two matrices.

4.9 Candidate CVD distortion models

Two candidate model classes were evaluated. Each parameterizes the angular distortion $\delta\theta(\theta)$ that maps the displayed hue angle θ to the angle perceived by a CVD observer.

Candidate Model 1: retinal-plus-gain (R+C).. The prevailing account attributes CVD color distortion to a cone-spectral shift at the retinal level (Machado et al., 2009), optionally augmented by a cortical compensation gain g (Boehm et al., 2014; Tregillus et al., 2021):

$$\delta\theta_{\text{RC}}(\theta; g) = (2 - g) \delta\theta_{\text{Mach}}(\theta) \quad (2)$$

where $\delta\theta_{\text{Mach}}$ is the per-hue angular shift predicted by Machado simulation applied to Stockman & Sharpe LMS cone fundamentals (Stockman & Sharpe, 2000). The gain g parameterizes cortical compensation: at $g = 1$ the cortical stage leaves the retinal distortion unchanged; at $g = 2$ it fully cancels the retinal shift; at $g > 2$ cortical processing reverses the direction of the retinal shift, moving the perceived hue past the undistorted color. The confusion axis is the color-space direction along which CVD observers cannot distinguish hues from one another. Because $\delta\theta_{\text{Mach}}$ lies entirely along this axis, R+C can only displace colors along this single direction and cannot account for distortions in other color directions.

Candidate Model 2: two-component cortical (2-comp). We propose a parametric model of the hue distortion pattern measured in §5.2 and §5.3:

$$\delta\theta(\theta; \beta_s, \beta_c) = \beta_s \cos(\theta - 90^\circ) + \beta_c \cos(\theta - \theta_{\text{conf}}) \quad (3)$$

where θ_{conf} is the CVD-subtype confusion axis (16° protan; 150° deutan), derived from Stockman & Sharpe cone fundamentals (Stockman & Sharpe, 2000). β_s parameterizes distortion along the S-cone axis ($\theta = 90^\circ$) and β_c parameterizes distortion along the individual confusion axis; these two components span the axes of distortion identified by the neural analysis (§5.2, §5.3). Both signs of β_c are unconstrained. We constrain $\beta_s \geq 0$ as a weak directional prior that fixes only the sign, not the magnitude, of the S-cone-axis term, consistent with reports that CVD perceptual errors concentrate near

the S-cone confusion loci Emery et al., 2021. Model adequacy was assessed by the procedure in §4.11; results are reported in §5.5.

4.10 Inverse fitting

Per-subject distortion parameters (β_s, β_c) were estimated by finding the distortion that, when applied to HC-derived neural representations, best reproduces the observed CVD responses. Three loss families served as the fitting objective.

Behavioral JND ratio (L_γ). For each color pair, the JND ratio $\gamma = \gamma_{\text{CVD}}/\bar{\gamma}_{\text{HC}}$ quantifies how much more difficult discrimination is for a CVD observer relative to the HC mean (§4.14). Two variants enter the fitting: a subtype-targeted pair ratio γ_{subtype} (the hue pair on each participant’s confusion axis predicted to show the largest distortion) and an aggregate ratio γ_{all} averaged across all tested pairs. L_γ penalises rank discordance between the model-predicted distortion magnitude at each pair (from Eq. 3) and the observed γ ordering.

Representational $\Delta\text{RDM cosine}$ (L_{RDM}). The between-group geometry difference $\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC}}$ captures which color pairs drive the CVD–HC representational deviation (§4.8). For a candidate distortion, $\Delta\text{RDM}_{\text{sim}}$ is obtained by applying $\delta\theta$ to the HC mean RDM. The loss is the cosine dissimilarity between the observed and simulated 28-element upper-triangle vectors:

$$L_{\text{RDM}} = \frac{1 - \cos(\Delta\text{RDM}_{\text{sim}}, \Delta\text{RDM}_{\text{obs}})}{2} \quad (4)$$

hV4 LOCO voxel-prediction (L_{LOCO}). The per-hue vulnerability profile $\mathbf{v}_{\text{obs}} \in [0, 1]^8$ records where each CVD participant’s hue interpolation breaks down (§4.7). For a candidate (β_s, β_c) , the simulated profile $\mathbf{v}_{\text{sim}}$ is obtained by re-applying the HC-trained ridge-GCV encoding weights W (§4.6) to the channel design matrix C shifted by the candidate distortion. The loss is the mean squared deviation between the two profiles:

$$L_{\text{LOCO}} = \frac{1}{8} \|\mathbf{v}_{\text{sim}} - \mathbf{v}_{\text{obs}}\|^2 \quad (5)$$

Grid search. Parameters were searched by exhaustive grid over $(\beta_s, \beta_c) \in [-90^\circ, 90^\circ]^2$ at 2° resolution (8,281 cells). For R+C (Eq. 2), g was searched over $[0, 3]$. Per-subject loss combinations and parameter selection are described in §4.11.

4.11 Parameter selection

Per-subject loss combinations and model class were selected by a three-gate procedure applied to results from a 5-train/2-test HC resample ($N = \langle 300 \rangle$) and a strict 7-fold HC leave-one-out (LOO) using the same atom factories.

Gate 1: Directional precondition. Only loss atoms where the CVD loss value exceeded the HC LOO distribution in the expected direction were admitted. Admission required signed Cohen’s $d \geq +0.5$; atoms failing this gate were excluded from combination.

Gate 2: Boundary saturation. Among admitted combinations, those where $\geq 50\%$ of resample solutions saturated the grid boundary were rejected as indicative of model misspecification.

Gate 3: Held-out test-loss. The primary criterion was the median composite test-loss ($\bar{L}_{\text{test}}$) across the $N = \langle 300 \rangle$ HC resamples. For each resample, the composite loss was evaluated at the training argmin on the two held-out HC, z-normalized relative to the training grid. Lower $\bar{L}_{\text{test}}$ indicates generalization to unseen HC references. The secondary criterion was the test-loss IQR across resamples, measuring resample stability of the generalization estimate. A dedicated neural RDM held-out loss ($\bar{L}_{\text{RDM}}^{\text{LOO}}$) from the strict 7-fold leave-one-out isolates the neural component’s held-out performance from the behavioral weighting in the composite. Supplementary stability indices included the IQR of recovered $(\hat{\beta}_s, \hat{\beta}_c)$ (parameter stability), the fraction of resamples recovering the modal 45° bin (mode share), and the strict 7-fold LOO parameter range. The HC false-positive rate was not itself used as a selection criterion; it is reported descriptively in §5.7.

4.12 Identifiability and recovery

Four pre-specified checks were run on each production candidate before any parameter was reported. (1) Parameter recovery: synthetic CVD data at a known ground truth were forward-projected through each HC donor’s encoder with spatially correlated voxel noise, and the full fitting pipeline was re-run ($n = 140$ per candidate); recovery was summarized by f_{10° , the fraction of refits within 10° of ground truth on both axes (pass $f_{10^\circ} \geq 0.5$, $|\text{bias}| < 10^\circ$). (2) Origin-recovery null: the same synthesis with ground truth at $(0^\circ, 0^\circ)$ and each donor’s real JND, recording the procedure’s per-axis uncertainty floor (excluded from the test pool, as it returns no p -value). (3) HC pseudo-CVD specificity: each HC participant’s real amplitudes were substituted into the CVD slot in turn, giving a seven-member null and a one-sided rank-distance percentile. (4) Color-label permutation: trial labels of the real CVD data were shuffled $N = 1000$ times with the HC reference held fixed, giving a one-sided p_{perm} . The three test-bearing checks across the two production candidates (six tests) were corrected together under the Benjamini–Hochberg false-discovery rate ($\alpha = 0.05$); outcomes are reported in §5.7.

4.13 Stimulus-space filter

The subject-specific correction filter was derived as the pre-image of the fitted 2-component transform $T(\hat{\beta}_s, \hat{\beta}_c) : \theta \mapsto \theta + \delta\theta(\theta; \hat{\beta}_s, \hat{\beta}_c)$. For each of the 8 displayed hues θ_k , we solved for the input angle $\tilde{\theta}_k$ such that $T(\tilde{\theta}_k) = \theta_k$, yielding a per-hue correction $\delta\theta_k^{\text{filt}} = \tilde{\theta}_k - \theta_k$. A CVD observer viewing $\tilde{\theta}_k$ thus receives, under the model, the same cortical hue representation as a healthy control observer viewing θ_k .

Pre-images were computed by numerical inversion of the forward map (Brent’s-method root-finding) to a convergence tolerance of $< 0.001^\circ$.

4.14 Behavioral tasks

Just-noticeable difference (JND).. Color discrimination thresholds were estimated for eight hue pairs using two-interleaved 1-up/1-down adaptive staircases, each terminating after eight reversals (Levitt, 1971). Pairs spanned three categories: (i)

three pairs hypothesized as universally difficult for both CVD types (yellow–purple, blue–purple, red–orange), (ii) two pairs probing the deutan confusion axis (orange–yellow, yellow–green), (iii) two pairs probing the protan confusion axis (cyan–magenta, green–blue), and (iv) one control pair (red–cyan). For each pair, the JND ratio $\gamma = \gamma_{\text{CVD}}/\bar{\gamma}_{\text{HC}}$ was computed, where $\bar{\gamma}_{\text{HC}}$ is the HC mean threshold. Ratios $\gamma > 1$ indicate elevated discrimination difficulty relative to HC. These ratios enter the fitting loss as described in §4.10.

Filter evaluation (the deutan participant). The deutan participant completed a second scanning session that compared the personalized filter and a deployed macOS accessibility filter against the unfiltered Session-1 baseline; the protan participant was not collected. The personalized filter was the per-subject 2-component stimulus-space pre-image, with parameters frozen from the main analysis and therefore out-of-sample. Because the deployed and personalized filters use different rendering pipelines, differences between them are interpreted as movement toward or away from the HC reference, not as evidence for a corrective mechanism. Acquisition, comparator, run-count, and single-case-inference details are given in Supplementary §8.

On the second-session data we recomputed, within each condition, the primary Session-1 interpolation metric (LOCO adjacent accuracy, §4.7), LORO classification, SRM Procrustes disparity into the HC shared space, and the representational dissimilarity matrices. As a secondary encoding index we also computed the forward-tuning correlation, the Spearman ρ between each ROI’s per-hue decoded hue and the canonical hue map. Because it indexes the encoding fit rather than the validated decoding readout, it is reported as corroboration only.

Each condition’s neural index was reported as a standardized single-case effect size (Cohen’s d against the HC distribution) rather than an inferential p , because the grand-mean permutation null is biased for this design. Behavioral testing (JND staircases and an 8AFC color-identification task) was repeated under both filters, each compared to the unfiltered Session-1 baseline and to the HC distribution; this interleaved, cross-session behavioral comparison is descriptive.

4.15 Reproducibility

All analyses were conducted in Python 3.10 (numpy 1.24.3, scipy 1.11.3, scikit-learn 1.3.0, BrainIAK 0.11). Random seeds were fixed (`seed=42`) for all permutation tests and bootstrap resampling. Analysis code and preprocessing scripts are available at https://github.com/Transconnectome/colorBlind_analysis. Anonymized preprocessed data are available on reasonable request to the corresponding author.

5 Results

5.1 Color discrimination is preserved across all ROIs in both CVD cases

A corrective stimulus-space filter is valid only if the visual cortex of a CVD individual retains categorical discrimination among the target hues. We verified this precondition using leave-one-run-out (LORO) cross-validation of the forward encoding model in SRM-aligned space.

Both CVD participants exceeded the 0.125 exact-accuracy chance level at every ROI (Figure 4A). A cross-subject generalization test of the same forward encoding model (HC-to-HC vs. HC-to-CVD pair-wise cross-decoding, all ROIs pooled) showed no HC-CVD difference (Mann-Whitney U test, 21 HC-to-HC vs. 14 HC-to-CVD pairs, $p = 0.668$; Supplementary §S14). The same null held at the within-ROI level at hV4 (Crawford & Howell t -test, both CVD subjects within the HC range, $p = 0.142$). The 8 scanner stimuli are therefore individually distinguishable at the cortical level in both CVD participants, in line with prior work on CVD above-threshold identification (Boehm et al., 2014; Bosten, 2019).

5.2 Color interpolation is reduced at hV4 in both CVD cases

We next tested whether continuous hue interpolation fails in CVD, using leave-one-color-out (LOCO) decoding of the forward encoding model. Among the four retinotopic ROIs, only hV4 supported above-chance hue interpolation in HC (adjacent accuracy 0.47 ± 0.05 , mean $\pm$ SEM over $n = 6$ HC, sub-07 excluded at hV4 for

insufficient voxels; $p = 0.008$ under 1,000 per-subject label permutations). V1–V3 did not significantly exceed chance despite preserving cross-run discrimination (V1 $p = 0.164$; V2 and V3 below chance). This is consistent with hV4 as the primary interpolation gate, replicating Brouwer and Heeger (2009).

Both CVD participants fell at or below the adjacent-accuracy chance level at hV4 (Figure 4B). The reduction was significant in the protan participant, whose adjacent accuracy of 0.13 fell below the HC distribution (Crawford & Howell $t = -2.91$, $p = 0.017$, $d_{cc} = -3.14$). In the deutan participant the reduction was a comparably large effect that did not reach significance (adjacent accuracy 0.25, $d_{cc} = -1.99$, $t = -1.84$, $p = 0.063$), in line with the limited power of a single-case test against $n = 6$ controls.

Descriptively, the reduction was not uniform across hues. Both participants showed zero adjacent accuracy at hV4 for the S-cone intermediate colors blue, purple, and magenta (Figure 4C). Per-hue single-case tests (Crawford & Howell, one-tailed, uncorrected and exploratory) reached significance at no individual hue against the $n = 6$ HC distribution. These per-hue accuracies form each participant’s hue vulnerability profile $\mathbf{v} \in [0, 1]^8$.

5.3 Color geometry is distorted at a distinct ROI in each CVD case

To characterize the geometric basis of the LOCO impairment, we compared the pairwise representational structure of CVD and HC participants in a shared SRM-aligned space. For each CVD participant we computed disparity and the per-pair RDM deviation from the HC mean ($\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC}}$).

The ROI of elevated disparity differs between the two CVD participants: V2 in the deutan participant and V1 in the protan participant (Figure 5B). The protan participant showed significantly elevated disparity specifically at V1 (Crawford & Howell $p = 0.007$ in the common HC space; $p = 0.045$ under the symmetric LOSO control), with no elevation at V2, V3, or hV4. The deutan participant showed elevated disparity at V2 (common-space $p = 0.040$) that attenuated to a non-significant trend under the symmetric LOSO control ($p = 0.116$), with no elevation at V1, V3, or hV4.

Full test statistics and effect sizes (t , d_{cc}) for both estimators are reported in table S1.

The Δ RDM heatmaps (Figure 5A) reveal subject-specific distortion structures. The deutan participant shows elevated distances among S-cone intermediate pairs at V2, and the protan participant shows a broader reorganization of pairwise distances at V1. Disparity tests the absolute amplitude of pairwise distances against the HC leave-one-out distribution, whereas the pairwise Δ RDM shows the directional pattern of those deviations; the disparity elevation — significant and LOSO-robust at V1 in the protan participant, a common-space trend at V2 in the deutan participant — is independent evidence that the cortical color representation is geometrically distorted.

5.4 The retinal-plus-gain model is structurally insufficient

The R+C model is structurally insufficient for both CVD participants, though the supporting evidence differs by case. For the deutan participant, $\langle 100\% \rangle$ of resample solutions saturated the grid boundary ($g = 3.0$), exceeding the 50% boundary-saturation gate (Section 4.11) and rejecting the model as misspecified (Wilson & Collins, 2019). For the protan participant, saturation was lower ($\langle 41\% \rangle$, $g = 2.95$) and did not by itself trip that gate; here the model is excluded instead by its fitted overcompensation ($g > 2$, below) and its degree-of-freedom deficit (below), and it is in any case dominated on held-out loss by the 2-component fit ($\bar{L}_{\text{test}} = -0.86$ vs -1.54 ; Section 5.5).

A gain $g > 2$ implies that cortical processing reverses the direction of the retinal shift past the undistorted hue (Section 4.9). Both participants scored below the Ishihara criterion (the deutan participant: 5/14 plates correct; the protan participant: 7/14), confirming residual CVD under standard viewing conditions. A fitted $g > 2$ is therefore internally inconsistent with confirmed CVD status: the model claims overcompensation where the participants demonstrably show impairment. Therefore, the saturated boundary is a model failure, not a valid parameter estimate.

The structural cause is a degree-of-freedom deficit. The function $\delta\theta = (2 - g) \cdot \delta\theta_{\text{Mach}}$ displaces hues only along the fixed confusion axis. No value of g can account for distortions in directions orthogonal to that axis. R+C is therefore

excluded as a filter candidate.

5.5 A common cortical model fits both CVD cases

The 2-component cortical model fit both participants using the same functional form. The deutan and protan cases differ in $(\hat{\beta}_s, \hat{\beta}_c)$, not in model structure. We fit the two-component parameters (β_s, β_c) for each participant and inverted the fitted transform to a stimulus-space filter.

For the deutan participant ($\theta_{\text{conf}} = 150^\circ$), the selected loss combination $\gamma_{\text{OY}} + L_{\text{RDM}}^{(V2)}$ yielded $(\hat{\beta}_s, \hat{\beta}_c) = (6^\circ, -42^\circ)$. The held-out test-loss was $\bar{L}_{\text{test}} = -2.36$ (IQR = 2.15), lower than the competing β_s -dominant candidate $(\hat{\beta}_s, \hat{\beta}_c) = (38^\circ, -10^\circ)$, which reached only $\bar{L}_{\text{test}} = -1.14$ and was therefore not adopted. The HC-resample parameter IQR was $(8^\circ, 2^\circ)$. The strict 7-fold LOO $\hat{\beta}_c$ range was $[-46^\circ, -38^\circ]$; all values are negative, confirming a stable sign across held-out reference sets.

For the protan participant ($\theta_{\text{conf}} = 16^\circ$), the selected loss combination $\gamma_{\text{all}} + L_{\text{RDM}}^{(V1)}$ yielded $(\hat{\beta}_s, \hat{\beta}_c) = (2^\circ, +24^\circ)$. The held-out test-loss was $\bar{L}_{\text{test}} = -1.54$ (IQR = 1.42), nearly tied with its closest grid competitor ($\bar{L}_{\text{test}} = -1.52$); the choice among neighbouring 2-component cells was therefore weakly determined, although both improved on the alternative retinal-plus-cortical model class scored on the same composite ($\bar{L}_{\text{test}} = -0.86$), locating the advantage in the model class rather than the precise grid cell. The HC-resample parameter IQR was $(0^\circ, 0^\circ)$, and $\langle 87.7\% \rangle$ of resamples recovered the same 45° bin. The strict 7-fold LOO IQR was $(0^\circ, 0^\circ)$. However, this stability is specific to the PCA-basis loss: the argmin shifts under the alternative SRM-basis reduction, so the recovered position is metric-dependent.

For neither participant did the LOCO loss family enter the selected combination; both winning combinations were JND-plus- Δ RDM.

The RDM atom in each subject’s winning combination corresponded to the ROI of elevated geometric disparity: V2 for the deutan participant (common-space $p = 0.040$; LOSO trend $p = 0.116$) and V1 for the protan participant ($p = 0.007$; LOSO $p = 0.045$, Section 5.3). The loss combination was selected by held-out test-loss. The

significant disparity ROI was identified separately, by Crawford–Howell tests on disparity amplitude. These are distinct procedures, but both read out the same per-ROI CVD–HC geometric deviation, so their agreement on ROI is expected and we do not treat it as independent corroboration.

On held-out folds, both fitted distortions reproduced the observed CVD geometry, beating the no-correction baseline on all 7 folds and ranking in the top 5–8% of grid combinations. The HC noise ceiling — the best fit achievable given HC data noise — was estimated from split-half HC amplitude reliability (Lage-Castellanos, Valente, Formisano & De Martino, 2019) and is conservative because CVD amplitude noise is excluded. Relative to this ceiling, the fits reached 52% (the deutan participant) and 67% (the protan participant) of the way from baseline to ceiling (held-out losses in table S2).

5.6 Neural data identifies what behavior cannot

Fitting behavioral (γ) and neural (RDM) loss atoms independently reveals three benefits of using the neural component.

First, the neural term captures a protan cortical direction that the behavioral loss cannot resolve. The protan participant’s behavioral-only fit yielded $\hat{\beta}_c \approx +4^\circ$ and did not beat the no-correction baseline ($\Delta L = +0.01$, 3/7 folds). The RDM term captures a signal that the behavioral loss cannot detect for this participant.

Second, the neural term stabilizes the argmin for the deutan participant. The behavioral-only and combined fits share the same argmin ($6^\circ, -42^\circ$), confirming that the behavioral atom independently locates the deutan direction; the neural-only fit independently corroborates this direction with an estimate of $\hat{\beta}_c = -26^\circ$. Adding the RDM atom reduced the boundary saturation rate from 23% to 9.3%, sharpening the argmin without shifting it.

Third, the neural term enhances parameter stability by reducing its IQR in both participants. For the deutan participant, adding the neural term reduced the HC-resample IQR from $(18^\circ, 6^\circ)$ to $(8^\circ, 2^\circ)$ (PCA) and $(10^\circ, 4^\circ)$ (SRM). For the protan participant, the IQR narrowed from $(6^\circ, 4^\circ)$ to $(0^\circ, 0^\circ)$ (PCA, deterministic) and $(0^\circ, 2^\circ)$

(SRM).

5.7 Identifiability: mechanism class recoverable, per-axis magnitude not

These fits recover the distortion’s mechanism class and the direction of its dominant axis, but not its precise per-axis magnitudes. Parameter recovery fails across all pre-specified identifiability checks (0/6 significant after FDR correction, BH $\alpha = 0.05$; Supplementary §S15). The voxel-level parameter-recovery test yielded $f_{10^\circ} < 0.30$ for both production candidates. The non-dominant axes ($|\hat{\beta}_s| \leq 6^\circ$ for both participants) fall below the effective parameter uncertainty of $\sim 20^\circ\text{--}25^\circ$ and are not recoverable (Supplementary §S15). The dominant axis ($|\hat{\beta}_c| = 42^\circ$ for the deutan participant) exceeds this uncertainty and is partially recoverable: when CVD responses are simulated from a known distortion and re-fitted, the recovered $\hat{\beta}_c$ deviates from the true value by 4.7° on average (Supplementary §S15).

Consistent with the partial recoverability of the dominant axis, the sign of $\hat{\beta}_c$ — the distortion direction — is stable across held-out reference sets (Section 5.5). The deutan participant consistently returns $\hat{\beta}_c < 0$ across all LOO folds and resamples. The protan participant consistently returns $\hat{\beta}_c > 0$ under the PCA-basis loss; sign stability under the SRM-basis loss was not verified (Supplementary §S15). The distortion direction is recoverable even when its magnitude is not.

The loss landscape shows where this magnitude indeterminacy lies. Although each participant’s selected fit is the global minimum on the full data (Figure 6), the per-resample argmins do not all concentrate there. For the protan participant, 87.7% recur at the selected fit, but a secondary basin near $(32^\circ, 0^\circ)$ marks a metric-dependent alternative. For the deutan participant, they disperse along the β_c axis, leaving its magnitude loosely constrained.

5.8 Per-subject stimulus-space filter

The per-subject filter $\delta\theta_k^{\text{filt}} = \tilde{\theta}_k - \theta_k$ is the pre-image of the fitted 2-component transform (Section 4.13).

For the deutan participant ($\hat{\beta}_s = 6^\circ$, $\hat{\beta}_c = -42^\circ$, $\theta_{\text{conf}} = 150^\circ$), the dominant $\hat{\beta}_c$ term produces a large-amplitude correction that varies asymmetrically across hues (mean $|\delta\theta| = 26.3^\circ$; Figure 7).

For the protan participant ($\hat{\beta}_s = 2^\circ$, $\hat{\beta}_c = +24^\circ$, $\theta_{\text{conf}} = 16^\circ$), the smaller $\hat{\beta}_c$ produces a more targeted correction concentrated near the protan confusion axis (mean $|\delta\theta| = 16.2^\circ$; Figure 7).

The two fits differ in the magnitude and direction of the dominant confusion-axis term ($\hat{\beta}_c = -42^\circ$ for the deutan participant vs. $+24^\circ$ for the protan participant).

5.9 Behavioral and neural filter evaluation

Neural. We evaluated the personalized filter in a second session in the deutan participant (the protan participant was not collected). The primary endpoint was the metric validated in Session 1: hV4 LOCO adjacent accuracy, the only ROI where HC interpolation exceeded chance. On this metric the personalized filter moved the participant partway toward the HC reference (baseline $0.23 \rightarrow$ personalized 0.31 ; HC 0.46 ± 0.11), whereas the deployed macOS accessibility filter did not (deployed 0.25 , indistinguishable from baseline). The personalized filter alone improved interpolation, but it remained below both the HC level and the 3/8 chance threshold ($d_{cc} = -1.35$ vs. HC; deployed $d_{cc} = -1.94$): the recovery was partial, not complete.

A secondary encoding index corroborated this ordering and separated the conditions more sharply. The forward-tuning correlation reached the HC level at hV4 under the personalized filter ($\rho = +0.18$ vs. HC $+0.21$). The deployed filter instead drove it negative ($\rho = -0.39$). This pattern held across all four ROIs (Fig. 8B; personalized ρ : V1 $+0.21$, V2 $+0.10$, V3 $+0.05$, V4 $+0.18$; deployed ρ : V1 -0.32 , V2 -0.19 , V3 -0.32 , V4 -0.39) and survived dropping any single run and re-extraction on the baseline voxel set ($\Delta\rho = 0.37\text{--}0.57$). Its V1 component attenuated and reversed under voxel matching ($d = +0.97 \rightarrow -0.82$), indicating partial dependence on occipital coverage. Because this index lies outside the validated Session-1 decoding pipeline, we report it as corroboration rather than as the primary endpoint.

Two further readouts separated what the filters preserved from what they did not restore. Color decodability survived both filters: LORO eight-way classification accuracy stayed near the HC level across conditions (≈ 0.69 – 0.72 at V1; HC 0.71), far above the 0.125 chance level, confirming that the second-session color signal was intact.

Representational geometry, by contrast, was not restored by either filter. SRM disparity favored the personalized over the deployed filter in all ROIs, but the no-filter baseline had the lowest disparity of the three (surviving run-count matching). The shared-space RDM agreed: neither filter was more HC-like than no filter by representational geometry (Fig. 8C,D). The personalized filter thus partially restored hue interpolation at hV4 while reshaping the color geometry into a configuration unlike both the unfiltered deutan and HC, and more compressed than either.

Behavioral. Both filters improved behavior. The baseline deutan deficit (JND mean $|z| = 2.24$, with the orange–yellow, yellow–green, and yellow–purple pairs all exceeding the HC empirical maximum) was eliminated by both the deployed filter ($|z| = 0.85$) and the personalized filter ($|z| = 0.78$). 8AFC color-identification accuracy rose from 0.81 at baseline to 0.97 under both, abolishing the yellow–green and purple–magenta confusions. The two filters could not be ranked behaviorally (Wilcoxon $p = 0.84$) and failed on different pairs (the personalized filter over-corrected orange–yellow below the HC range). The single-subject, single-run, cross-session design with a shared-pipeline confound precludes a behavioral ranking. Both a well-configured deployed filter and the personalized filter thus restore the deutan participant’s discrimination to near-control levels. On the validated neural metric, the personalized filter’s advantage is a partial move of hV4 hue interpolation toward the HC reference that the deployed filter does not produce; the encoding forward-tuning index corroborates this.

6 Discussion

We asked whether the cortical color representation of a color-vision-deficient (CVD) individual is structured enough to ground a personalized corrective filter. Two

CVD participants, one deutan and one protan, retained categorical discrimination of all eight color hues at every retinotopic ROI. Yet both selectively lost continuous-hue interpolation, falling at or below the adjacent-accuracy chance level. This deficit was not diffuse signal loss but a structured geometric distortion. We modeled it with two cortical components: a rotation along the subtype confusion axis — the classical direction of CVD color confusion — and a distortion along the S-cone axis, where the interpolation deficit was concentrated. Fit independently per participant from behavioral discrimination thresholds and cortical representational geometry, this model recovered an exact stimulus-space pre-image for all eight hues in both cases. That pre-image is the per-person correction filter, built from that individual’s own cortical color geometry. We show, in the deutan case, that such a filter partially moves the cortical hue interpolation toward the healthy-control reference, without reaching it. Behavioral discrimination was restored, but not beyond a deployed accessibility filter.

6.1 A geometric distortion of cortical color representation

In both CVD participants, the deficit appeared as a structured distortion of cortical color geometry rather than a uniform attenuation of the chromatic signal. Representational geometry treats the pairwise distance structure among stimuli as an encoded relationship in each cortical region (Kriegeskorte, Mur & Bandettini, 2008; Kriegeskorte & Douglas, 2018). Two complementary measurements expose a structural distortion, each at the stage where its signal exists. The representational dissimilarity matrix (RDM) shows *which* pairwise distances deviate from the healthy-control (HC) geometry: disparity was significantly elevated at V2 in the deutan participant (Crawford & Howell $p = 0.040$) and at V1 in the protan participant ($p = 0.007$). The two participants showed this elevated disparity at different areas (V2 and V1), not a common one. Leave-one-color-out decoding shows *where* the continuous-hue manifold breaks down. Only hV4 supported above-chance interpolation in HC (0.47 ± 0.05 , $p = 0.008$), replicating Brouwer and Heeger (2009). There, the protan participant fell significantly below HC (0.13, $p = 0.017$, exploratory single-case), and the deutan

participant showed a non-significant reduction in the same direction (0.25 , $p = 0.063$). Both sat at zero adjacent accuracy at the S-cone intermediate hues blue, purple, and magenta. No individual hue reached significance in uncorrected, exploratory per-hue tests. This account aligns with prior characterizations of CVD as a multidimensional deformation of color space rather than a uniform attenuation (Boehm et al., 2014; Emery et al., 2021).

6.2 A neurally grounded, individualizable correction filter

We built each correction filter by inverting that participant’s fitted two-component model to its stimulus-space pre-image (Methods §4.13) — the displayed hue that, after the cortex distorts it, is perceived as the intended color. Across the eight hues this filter applied a mean rotation of $|\delta\theta| = 26.3^\circ$ for the deutan participant and 16.2° for the protan participant.

Fitting the behavioral and neural loss atoms independently revealed three benefits of grounding the correction in cortical geometry. First, the neural term recovered a confusion-axis direction the behavioral loss could not resolve: the protan behavioral-only fit did not beat the no-correction baseline ($\Delta L = +0.01$, 3 of 7 folds), whereas the RDM term resolved a clear rotation. Second, it stabilized an already-informative solution: for the deutan participant the behavioral-only and combined fits shared the same argmin ($6^\circ, -42^\circ$), and adding the neural term sharpened it without shifting it, lowering the boundary-saturation rate from 23% to 9.3%. Third, it tightened parameter stability in both participants, narrowing the resample interquartile range of the recovered parameters. The neural-only fit for the deutan participant independently returned the same direction ($\hat{\beta}_c = -26^\circ$), corroborating the behavioral estimate. A correction grounded in individual cortical geometry therefore differs in substance from a purely behavioral one.

The resulting filter is per-person and neurally grounded, the architectural advance over a population-average correction. The architecture is individualizable because each filter is built from one participant’s measured cortical color geometry rather than from a

fixed subtype-average spectral shift.

The two fitted filters differ by subtype in the *direction* of the dominant confusion-axis term ($\hat{\beta}_c = -42^\circ$ for the deutan participant versus $+24^\circ$ for the protan participant). Because θ_{conf} is subtype-defined and the sample holds one deutan and one protan, this contrast is between-subtype. It rests on the recoverable features of the fit: across held-out reference sets, the two-component mechanism class and the sign of the dominant confusion-axis term are stable, whereas the per-axis magnitudes are not separately identifiable (0 of 6 recovery checks survive FDR correction). The contrast therefore rests on these stable directions, not on absolute parameter values.

A population-average transform does not substitute for this per-person inverse, for a reason that holds independent of any fit. The Machado class — a fixed spectral shift along the single confusion axis, even when scaled by a cortical gain (Machado et al., 2009) — stays confined to that one axis and therefore does not represent distortion in orthogonal directions. For the protan participant it even compresses three distinct displayed hues into a narrow arc, leaving no exact pre-image where that participant is most vulnerable (Appendix A). Fitting the class confirmed the mismatch: its gain saturated near $g \approx 3$, implausibly implying overcompensation in observers who remain measurably impaired. But the structural limit alone rules it out as an exact correction (Section 4.9).

6.3 Filter evaluation in the deutan case

We evaluated the personalized filter against a deployed macOS accessibility filter in a second session with the deutan participant (the protan participant was not collected). On the validated interpolation metric (hV4 adjacent accuracy), the individualized filter moved the deutan participant partway toward the healthy-control reference, whereas the deployed filter did not improve on the unfiltered baseline; on a secondary encoding index the deployed filter additionally drove the hue map negative. Categorical discrimination (LORO) stayed intact under both filters, confirming that the cortical color signal in CVD is preserved and that neither filter degraded it.

The personalized filter did not, however, reproduce the full healthy-control color geometry. It reshaped the eight-hue dissimilarity structure (RDM) into a configuration unlike both the baseline and the controls. On that full-geometry measure the unfiltered baseline was in fact closest to the controls, because it presents the same physical colors, so RDM similarity is not the bar a corrective filter must clear. Behaviorally, both filters restored the participant’s hue discrimination to near-control levels and could not be ranked against each other (Wilcoxon $p = 0.84$). This evaluation rests on a single deutan case. Whether the personalized filter also outperforms the deployed filter behaviorally, and whether the result generalizes beyond this individual, remain open.

6.4 Limitations

Four considerations bound the proof-of-concept scope of these findings. First, the CVD sample is $N = 2$. Single-case statistics (Crawford & Howell, 1998) permit individual-level inference, but population-level claims about deutan or protan subtypes require replication with more participants. As established above, the sign of $\hat{\beta}_c$ is the recoverable quantity. Whether the between-subtype $\hat{\beta}_c$ difference (-42° vs. $+24^\circ$) is subtype-level or idiosyncratic awaits power-adequate replication. Establishing per-person rather than subtype-average correction specifically requires testing several individuals *within* a single subtype — the decisive follow-up experiment. Second, the stimulus set was confined to a single isoluminant, iso-chroma locus ($L^* = 75$, chroma = 40). The filter is therefore derived only along the hue circle at fixed lightness and saturation, and any future validation is similarly confined. It does not constrain how the correction generalizes to luminance- or saturation-varying stimuli or to naturalistic viewing — gaps that denser stimulus sampling across these axes would close. Third, the HC reference cohort is small (seven controls), so the HC $\|\hat{\beta}\|$ leave-one-out distribution we report is a magnitude anchor for the CVD estimates, not a hypothesis test. A larger HC reference cohort would be required to arbitrate population-level distortion patterns. Fourth, we did not compute parameter-level bootstrap confidence intervals on $\hat{\beta}_s$ and $\hat{\beta}_c$, so the per-axis estimates are point values whose interval precision awaits the

additional runs per participant that bootstrap resampling would require.

6.5 Conclusion

This work is a proof-of-concept that each individual’s cortical color geometry, measured with fMRI, can ground a stimulus-space correction obtained by inverting a fitted model of that individual’s representational distortion. It illustrates, in two cases, that a color-vision deficiency need not present as a uniform retinal loss undone by a fixed subtype-average spectral shift. It can instead be an individually patterned, multidimensional distortion of the cortical color representation — one structured enough to invert. Because the correction is the inverse of each individual’s own fitted distortion, it is specific to that person rather than a fixed, population-average retinal transform. Only the distortion’s form and dominant direction are recoverable, not its exact magnitude. In the deutan case, this correction partially moved cortical hue interpolation toward the healthy-control reference — further than the deployed filter, though still short of the HC level. A behavioral advantage and broader validation remain to be established. A correction spanning hue, lightness, and chroma jointly — fitted from denser stimulus sampling, more runs per participant, and a larger HC reference pool — is the path toward a deployable personalized filter. More broadly, treating a sensory deficit as an invertible distortion of its own cortical representation offers a template for personalized, brain-based perceptual correction beyond color vision.

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Appendix A: Alternative Model Comparison

The main text adopts a 2-component cortical model (Section 5.5). Here we compare it against the retinal-family alternative on the same held-out composite criterion (§4.11), and give the structural reasons the retinal-family class is inadequate for the stimulus-space filter task.

A.1 Retinal-plus-cortical (R+C) fit on the composite criterion

The retinal-plus-cortical model applies the subtype Machado cone-shift $\delta\theta_{\text{Mach}}$ Machado et al., 2009 scaled by a single cortical-gain parameter g ($\delta\theta_{\text{RC}} = (2 - g) \delta\theta_{\text{Mach}}$; $g = 1$ retinal-only, $g = 2$ exact cortical cancellation; Tregillus et al., 2021). Scored on the same held-out composite as the 2-component model, R+C failed for both participants: for the deutan participant every resample saturated the gain boundary ($g \rightarrow 3.0$, $\langle 100\% \rangle$), so the fit was rejected as misspecified; for the protan participant the best fit ($g = 2.95$) generalized worse than the adopted 2-component model ($\bar{L}_{\text{test}} = -0.86$ vs. -1.54). Full fits are in Supplementary Table S2 and Section 5.4. A gain $g > 2$ implies cortical overshoot past the undistorted hue, inconsistent with the participants' confirmed residual CVD, so the saturated estimate is a model failure rather than a parameter value.

A.2 Inverse-mapping (pre-image) behavior

The angular form of the 2-component model is bijective by construction: a stimulus-space pre-image $\theta + \delta\theta$ exists for any target hue. The retinal-shift component $\delta\theta_{\text{Mach}}$ (the $g = 1$ retinal-only case), by contrast, compresses three hues (green 135° , cyan 180° , blue 225°) into a narrow $\sim 26^\circ$ band (the eight shifted hues span only $\sim 96^\circ$) at the protan participant's subtype severity, so they lose a unique pre-image and exact inversion fails at those positions. This arc-compression is a structural property of the retinal-shift form, independent of the fitting criterion, and disqualifies the retinal-family baseline for the stimulus-space filter task of Section 5.8.

A.3 Why the cortical gain cannot recover the distortion

Because the cortical gain g rescales the retinal shift uniformly without generating the angular distortion captured by the 2-component model's $\beta_c \cos(\theta - \theta_{\text{conf}})$ term, no value of g matches the observed hV4 LOCO vulnerability pattern beyond what the retinal $\delta\theta_{\text{Mach}}$ term alone provides. This single-axis degree-of-freedom deficit is the structural cause of the failure in A.1, and it holds across fitting criteria.

8 Supplementary Methods

S1. Confound Regression and Temporal Filtering

Within-run head motion was estimated via rigid-body realignment. No temporal filtering or confound regression was applied; slow drift was modeled via linear per-run regressors in the general linear model.

S2. Quality Control

Registration quality was evaluated by computing the intersection between the Wang Atlas ROIs (V1, V2, V3, hV4) and the BOLD brain mask in MNI space. Across 10 subjects, mean ROI coverage was 84.3% (SD = 21.7%), and GLM valid ratio (percentage of ROI voxels with reliable stimulus-evoked responses) was 99.6%. We evaluated all subjects' data suitable for downstream analysis, though sub-07 showed reduced coverage (30.8%) due to individual anatomical variability.

S3. Mean Activation Analysis

To verify that the shared-space geometric differences are not a signal-quality confound, we compared activation-level metrics between groups before SRM alignment. Mean activation magnitude (mean $|\beta|$), signal-to-noise ratio, color modulation depth, run-to-run reliability, and spatial variance did not differ between HC and CVD across V1, V2, V3, and hV4 (Welch's t -tests, all $p > 0.09$), and the two CVD cases fell within the HC range on every metric (Crawford–Howell, all $p > 0.09$). Voxel-level color selectivity was strong in both groups (one-way ANOVA across voxels, $F > 4$, $p < 0.001$ in V1–V3). Critically, none of the activation metrics correlated with SRM disparity across participants (Pearson r , 20 tests, all $p > 0.13$). The HC–CVD difference is therefore one of representational pattern, not signal amplitude: the cortical color response is equally strong in CVD but geometrically reorganized.

S4. Statistical Analysis

Unless noted otherwise, the significance threshold was $\alpha = 0.05$. False Discovery Rate correction (Benjamini & Hochberg, 1995) was applied within two pre-specified families only: the RDM color-pair comparisons ($q = 0.05$; 28 pairs per participant) and the six parameter-identifiability checks (section 4.12). The exploratory per-hue vulnerability comparisons and the per-pair identification tests (including the blue-hue deficit) are reported one-tailed and uncorrected. Individual CVD participants were compared to the HC distribution using Crawford and Howell (1998) modified t-tests:

$$t^* = \frac{x_{\text{CVD}} - \bar{x}_{\text{HC}}}{\text{SD}_{\text{HC}} \times \sqrt{(n+1)/n}}, \quad df = n - 1 = 6 \quad (6)$$

with one-tailed p-values testing the directional hypothesis that CVD disparity exceeds HC. Effect sizes were quantified via Hedges' g . Group-level permutation tests (10,000 iterations) were used for HC-CVD comparisons by permuting subject labels; color-level permutation tests (1,000 iterations) were used for LOCO/LORO validation by permuting color labels within subjects.

S5. Dimensionality Selection (K-Selection)

The reduced dimension k for SRM was determined via leave-one-subject-out (LOSO) cross-validation based on within-subject representational dissimilarity matrix (RDM) reliability (Cunningham & Yu, 2014). For each candidate k (range: 1–8) and fold, reliability was quantified as the Spearman correlation between RDMs computed from even runs (2,4,6) versus odd runs (1,3,5) of the held-out participant in SRM space. We selected k maximizing mean reliability across seven folds. The selected values and their associated cross-subject RDM Spearman correlations were:

- V1: $k = 4$ (cross-subject RDM correlation = 0.60)
- V2: $k = 4$ (cross-subject RDM correlation = 0.57)
- V3: $k = 3$ (cross-subject RDM correlation = 0.55)
- hV4: $k = 3$ (cross-subject RDM correlation = 0.32)

S6. Generalized Cross-Validation for Ridge Regression

The regularization parameter α for ridge regression was selected via Generalized Cross-Validation (GCV; Golub et al., 1979), implemented as `sklearn.linear_model.RidgeCV` with `gcv_mode='auto'` (Pedregosa et al., 2011). GCV provides a computationally efficient approximation to leave-one-out cross-validation for selecting α . The optimal α minimizes the GCV score:

$$\text{GCV}(\alpha) = \frac{1}{n} \|y - \hat{y}\|_2^2 / \left(1 - \frac{\text{tr}(H)}{n}\right)^2 \quad (7)$$

where X denotes the design matrix, y is the observed response vector, $\hat{y}$ is the ridge-predicted response, $H = X(X^T X + \alpha I)^{-1} X^T$ is the hat matrix, and n is the number of training samples.

S7. Leave-One-Out Consistent Disparity Estimation

To quantify HC-CVD disparity with unbiased references, we used a leave-one-out (LOO) consistent procedure: for each fold i (leaving out HC participant i), a reference pattern R_{-i} was computed as the mean of the remaining six HC participants' aligned data. Both the held-out HC_i and both CVD participants were compared against this same reference R_{-i} , yielding disparity scores $d(\text{HC}_i, R_{-i})$ and $d(\text{CVD}_j, R_{-i})$. This ensures HC and CVD disparities are measured against identical references within each fold. Individual CVD disparity significance was assessed via Crawford and Howell (1998) modified t-tests comparing each CVD participant's mean disparity (averaged across 7 folds) to the HC distribution ($n = 7$, $df = 6$).

Common-space vs. symmetric-LOSO estimates. The geometric-disparity tests (§5.3) are reported primarily in the common HC-trained SRM space, in which the seven HC train the shared space and each subject of interest is projected in; because the held-out HC reference is in-sample whereas the projected CVD participant is out-of-sample, we additionally computed a symmetric leave-one-subject-out (LOSO) estimate in which every HC and CVD subject alike is projected by SVD into a space trained on the remaining six HC (Methods, §4.8). Both use the Crawford & Howell,

1998 one-tailed single-case t ($df = 6$). Table S1 reports both estimates for all ROIs. The protan participant’s V1 elevation is significant under both estimators; the deutan participant’s V2 elevation is significant in the common space and attenuates to a non-significant trend under LOSO.

S8. Cross-Validation Procedures

Leave-One-Run-Out (LORO)

LORO evaluated whether individual colors could be distinguished from voxel response amplitudes, testing the consistency of neural color encoding across runs. For each of six runs, the corresponding response matrices (B , C) were held out as the test set, W was trained on the remaining five runs. The held-out run’s color patterns were then decoded by nearest-neighbour matching in channel-output space using the OLS pseudoinverse weights (Methods §4.6).

Leave-One-Color-Out (LOCO)

LOCO tested whether geometric relationships of circular hue space were preserved in neural representation by evaluating interpolation to held-out colors. For each color, all six run response matrices of the target color were held out. W was trained on the remaining seven colors. The held-out color was then decoded from voxel responses using the OLS pseudoinverse weights, yielding the adjacent-accuracy vulnerability profile. The complementary voxel-response prediction (ridge-GCV encoding) enters only the fitting loss L_{LOCO} , not the decoding metric (Methods §4.6).

Cross-run alignment and leakage control

The run-level Procrustes alignment (Methods, ROI definition and response estimation) maps runs 2–6 onto a fixed run-1 reference by an orthogonal rotation that uses no stimulus labels. Because the alignment target is a single fixed run rather than a jointly fitted shared space, the held-out run or color does not define the alignment frame. We further confirmed that re-estimating the Procrustes rotation within each

cross-validation fold (nested alignment) leaves LORO classification accuracy essentially unchanged, ruling out test-set leakage from the alignment step.

Leave-One-Subject-Out (LOSO)

LOSO evaluated zero-shot generalization to held-out participants. As LOSO requires cross-subject prediction, SRM-transformed data were used to align voxel spaces across participants. For each HC participant, W was trained on the remaining HC participants' shared-space data and tested on the held-out participant. This assessed whether the HC group-derived common space supported generalization to novel individuals.

S9. Evaluation Metrics

Color Decoding Metrics

Color decoding was evaluated using (1) mean absolute error (MAE) between predicted and true hue angles (chance = 90° , expected from a uniform random predictor on a circle), and (2) categorization accuracy with $\pm 22.5^\circ$ bins around each of the eight colors (chance = $12.5\% = 1/8$).

Voxel Prediction Metrics

Voxel response prediction was evaluated using Pearson correlation between predicted and observed response vectors, computed across all voxels and colors in the test set.

Statistical Significance

Statistical significance of group comparisons (HC vs CVD) and individual CVD analyses was assessed using nonparametric permutation tests (10,000 iterations; Nichols & Holmes, 2002).

S10. Shared Response Model Residuals

Each participant’s residuals in the SRM model reflect the extent to which their data are not captured by the common structure, including inherent geometric differences. The orthogonal constraint ($W_i^T W_i = I_k$) helps preserve geometric relationships between colors—distances and angles—in the common space.

Additionally, global between-subject RDM similarity was quantified using Spearman correlation across all 28 color pairs.

S11. Image Orientation Initialization

Image orientation was first initialized according to the scanner-defined obliquity information in the NIfTI header. The mutual-information cost function was then optimized to maximize the statistical dependence between T1w and BOLD intensity distribution within the overlapping region.

S12. Comparison with Retinal-Family Distortion Models

The retinal-family (R+C) model was evaluated alongside the 2-component model under the same grid-search and held-out composite criterion (§4.11).

Model. The retinal-plus-cortical (R+C) model is defined in §4.9: the subtype Machado cone-shift $\delta\theta_{\text{Mach}}$ Machado et al., 2009, fixed by the confusion axis, scaled by a single cortical-gain parameter g ($\delta\theta_{\text{RC}} = (2 - g) \delta\theta_{\text{Mach}}$; $g = 1$ is retinal-only, $g = 2$ exact cortical cancellation; Tregillus et al., 2021). It was fit by the same grid-search and held-out composite criterion as the 2-component model (§4.11), so the two model classes are directly comparable on $\bar{L}_{\text{test}}$.

Fits (Table S2). On the held-out composite, R+C failed for both participants. For the deutan participant, $\langle 100\% \rangle$ of resample solutions saturated the gain boundary ($g \rightarrow 3.0$), so the fit was rejected at the boundary-saturation gate as model misspecification (§5.4). For the protan participant, R+C passed that gate ($\langle 41\% \rangle$ saturation, $g = 2.95$) but generalized worse than the adopted 2-component fit ($\bar{L}_{\text{test}} = -0.86$ vs. -1.54). A gain $g > 2$ implies cortical overshoot past the undistorted

hue, inconsistent with the participants' confirmed residual CVD; the structural cause is the single-axis degree-of-freedom deficit detailed in §5.4.

Relation to Δ RDM and LOCO. Optimizing a retinal-family model directly against Δ RDM also fails for both CVD participants. For the adopted 2-component model, the Δ RDM cosine (L_{RDM}) is itself a component of the selected composite loss (§4.11); the forward-encoding LOCO family did not enter either adopted combination and is used only for diagnostic decoding.

S13. HC leave-one-out magnitude-anchor distribution

Because the production loss differs between participants (the deutan participant: $\gamma_{\text{OY}} + L_{\text{RDM}}^{(V2)}$; the protan participant: $\gamma_{\text{all}} + L_{\text{RDM}}^{(V1)}$), the HC leave-one-out anchor was computed per participant under that participant's loss. Each of the $n = 7$ HC participants was refit as a pseudo-CVD case with the remaining HC pool as reference, yielding a distribution of fitted-parameter magnitudes $\|\hat{\beta}\|$: 30.5° to 58.1° (mean 49.1°) under the deutan participant's loss, and 23.4° to 55.5° (mean 35.7°) under the protan participant's loss. Both CVD estimates fall within their respective HC ranges (the deutan participant $\|\hat{\beta}\| = 42.4^\circ$; the protan participant $\|\hat{\beta}\| = 24.1^\circ$), so parameter magnitude alone does not separate the CVD cases from HC; the distribution is reported as a descriptive magnitude anchor, not as a hypothesis test of CVD-versus-HC specificity.

S14. Effect Sizes for Single-Case Comparisons

Crawford–Howell single-case comparisons used the modified t Crawford & Howell, 1998; the corresponding case-control effect size is $d_{cc} = (x_{\text{case}} - \bar{x}_{\text{HC}})/s_{\text{HC}} = t \cdot \sqrt{(n+1)/n}$ where n is the HC sample size used in the test. Mann–Whitney U tests are reported with rank-biserial correlation r_{rb} as effect size. Effect sizes for the single-case interpolation comparisons are collected in table S3; the geometric-disparity effect sizes are listed alongside their t statistics in table S1 (case-control $d_{cc} = t\sqrt{8/7}$, $n = 7$). Two pooled cross-decoding comparisons were null and are reported by their p -values only: the HC-to-HC versus HC-to-CVD

generalization test (Mann–Whitney U , 21 vs. 14 pairs, $p = 0.668$) and the within-ROI hV4 single-case test ($p = 0.142$).

Supplementary S15: Identifiability checks

Four pre-specified checks assessed whether the production argmin is identifiable (Test 1), what the algorithm’s noise floor is (Test 2a), and whether the fit is specific to CVD data (Tests 2b–2c). Each test was run on the two production candidates (S08-robust, S09-primary). All tests use PCA-basis voxel synthesis (spatial PCA(20) on SRM-projected amplitudes) as the feature space. FDR correction (Benjamini–Hochberg, $\alpha = 0.05$) was applied across 6 tests (2 candidates $\times$ 3 main tests); 0/6 reached significance.

Test 1 — Voxel-level parameter recovery. Synthetic fMRI data were generated at the production argmin (PCA-basis; 7 HC donors $\times$ 20 noise draws = 140 samples per candidate) with GT-consistent JND: behavioral thresholds were scaled to match the distortion magnitude at the ground-truth (β_s, β_c) . The same fitting pipeline was re-run on each synthetic sample. Pass criterion: fraction within 10° on both axes ($f_{10^\circ} \geq 0.5$ and $|\text{bias}| < 10^\circ$ on both axes).

Test 2a — Origin-recovery null. Synthetic fMRI data were generated at $\text{GT} = (0, 0)$ using real HC JND (no synthetic behavioral signal; PCA-basis). At the null, real HC JND carries no CVD-consistent signal, making this a noise floor measurement free of synthesis design contamination. Pass criterion: $|\text{bias}| < 5^\circ$ on both axes.

Test 2b — HC pseudo-CVD specificity. Each HC participant’s real amplitude data were substituted as a fake CVD participant (7 carriers per candidate, deterministic). The production CVD fit distance from the origin was ranked among the 7 HC pseudo-CVD distances ($\text{rank}_{\text{dist}}$, one-sided high). Pass criterion: $\text{rank} = 1.0$ (CVD fit exceeds all HC distances). This test is descriptive only and was not used as a selection criterion.

Test 2c — Color-label permutation. CVD trial labels were randomly shuffled ($N = 1000$ permutations); HC data were held fixed. Pass criterion: $p_{\text{perm}} < 0.05$ (one-sided, lower = better).

FDR summary and basis caveat. After FDR correction across 6 tests (2 candidates $\times$ 3 main tests; BH $\alpha = 0.05$), no test reaches significance. The production argmin is treated as a descriptive embedding of the CVD-specific distortion direction throughout.

All results above use the PCA-basis loss. For the protan participant, the sign of $\hat{\beta}_c$ (mechanism class) was confirmed under the PCA-basis loss only. The SRM-basis loss may yield a different mechanism class for this participant; sign stability under the SRM basis was not verified.

Supplementary S16: Filter-evaluation session — design and comparator

Comparator. The deployed comparator was the macOS accessibility Color Filter (System Settings > Accessibility > Display, build 26.5.1), set at an intensity the participant self-tuned before scanning. It is the unmodified shipping product — an operating-system-level post-display transform. We compared against this deployed filter rather than a re-implemented retinal-simulation transform so that the comparator is the correction an end user actually encounters. The personalized filter, by contrast, was rendered directly in PsychoPy from the frozen per-subject pre-image.

Acquisition. Eight fMRI runs were acquired in ABBA-counterbalanced order, split evenly into four runs per filter. Scanner time set this count: a within-session two-condition design at six runs per condition would run roughly 90 minutes, beyond the 60–75-minute window over which participant attention and head motion stay acceptable. Four runs per condition (~ 60 minutes) sits at the recommended boundary.

Run-count adequacy. A subsampling analysis on the six-run Session-1 data — evaluating signal-quality, geometric-stability, and outcome-detection metrics across all $\binom{6}{n}$ run subsets for $n = 2-6$ — indicated that four runs retains the single-subject landmark findings, with five preferred and four the acceptable fallback. We extended the same subsampling to the primary interpolation metric. Across all $\binom{6}{n}$ subsets, hV4 LOCO adjacent accuracy retained the Session-1 landmark at every run count down to four (Figure 9): the healthy-control mean stayed above the $3/8$ chance level (at $n = 4$, 0.45 versus 0.46 at six runs), while both CVD participants stayed well below it (deutan 0.23, protan 0.14; single-case $d_{cc} < -2$ in both). The metric reported in the main text is therefore stable at the deployed run count. Four runs is, separately, below the count that simulations recommend for fine-grained RDM-based model discrimination, so the second session is powered to detect movement toward or away from the HC reference, not to adjudicate model classes. For every neural index, the Session-1 HC reference runs were subsampled to four to match the per-condition run count, so condition and HC estimates rest on equal data.

Single-case inference. Each condition’s neural index was compared to the HC distribution as a standardized single-case effect size (Cohen’s d against the HC mean and SD), reported in place of an inferential p because the grand-mean permutation null is biased for this design. Robustness was assessed in two ways: by leave-one-run separation (the contrast holding after any single run is dropped) and by re-extracting both conditions on the no-filter baseline voxel set.

Behavioral battery. JND staircases were run for eight hue pairs, and the eight-alternative forced-choice (8AFC) color-identification task comprised 64 trials. Both were compared to the unfiltered Session-1 baseline and to the HC distribution.

Table S1

Mean pairwise disparity, Crawford–Howell one-tailed single-case t ($df = 6$) against the HC reference, in the primary common HC-trained SRM space and under the symmetric LOSO control, with the common-space case-control effect size $d_{cc} = t\sqrt{8/7}$ ($n = 7$).

Bold: $p < 0.05$.

Subject	ROI	Common space			Symmetric LOSO	
		t	p	d_{cc}	t	p
Deutan	V1	1.10	0.157	1.18	0.48	0.323
	V2	2.11	0.040	2.26	1.33	0.116
	V3	1.92	0.052	2.05	1.17	0.143
	hV4	0.23	0.411	0.25	0.07	0.474
Protan	V1	3.48	0.007	3.72	2.02	0.045
	V2	0.99	0.181	1.06	0.77	0.234
	V3	0.09	0.466	0.10	0.06	0.479
	hV4	1.13	0.150	1.21	0.80	0.228

Table S2

Distortion-model fits for both CVD participants, all scored on the same held-out composite test-loss ($\bar{L}_{\text{test}}$, lower is better; §4.11; the neural RDM atom in each adopted fit is V2 for the deutan and V1 for the protan participant). The 2-component model is shown at its adopted fit. The retinal-plus-cortical (R+C) fits are reported for comparison: the deutan R+C was rejected at the boundary-saturation gate (§5.4), and the protan R+C generalized worse than the 2-component fit.

Participant	Model	Fitted parameters	Held-out fit
Deutan	2-component	$\beta_s = 6^\circ$, $\beta_c = -42^\circ$	$\bar{L}_{\text{test}} = -2.36$
Protan	2-component	$\beta_s = 2^\circ$, $\beta_c = +24^\circ$	$\bar{L}_{\text{test}} = -1.54$
Deutan	R+C	$g \rightarrow 3.0$ ($\langle 100\% \rangle$ saturated)	rejected (Gate 2)
Protan	R+C	$g = 2.95$ ($\langle 41\% \rangle$ saturated)	$\bar{L}_{\text{test}} = -0.86$

Table S3

Effect sizes for the single-case hue-interpolation comparisons at hV4 (main text, §5.2). Crawford–Howell case-control effect size $d_{cc} = t\sqrt{(n+1)/n}$ against the $n = 6$ HC distribution. Per-hue tests are one-tailed, uncorrected, and exploratory; both CVD participants share identical per-hue statistics where adjacent accuracy is zero. No individual hue reaches significance.

Comparison (hV4)	t	p	d_{cc}
LOCO adjacent accuracy, deutan	−1.84	0.063	−1.99
LOCO adjacent accuracy, protan	−2.91	0.017	−3.14
<i>Per-hue adjacent accuracy (one-tailed, exploratory; both participants)</i>			
Blue	−1.73	0.072	−1.87
Purple	−0.90	0.205	−0.97
Magenta	−1.32	0.122	−1.43

Candidate	bias (β_s, β_c)	f_{10°	Verdict
S08-robust ($6^\circ, -42^\circ$)	($+16^\circ, -4.7^\circ$)	0.26	FAIL
S09-primary ($2^\circ, +24^\circ$)	($+11^\circ, -27^\circ$)	0.14	FAIL

Table S4

Test 1 results. Both production candidates fail the recovery criterion ($f_{10^\circ} < 0.5$). The non-dominant axes ($|\hat{\beta}_s| \leq 6^\circ$ for both participants) fall below the effective parameter uncertainty established by Test 2a and are not recoverable. For S08-robust, the dominant axis ($|\hat{\beta}_c| = 42^\circ$, bias = 4.7°) shows lower bias than the non-dominant axis (16°), consistent with partial magnitude recoverability where the axis magnitude exceeds the noise floor.

Candidate	$ \hat{\beta}_s _{\text{med}}$ (IQR)	$ \hat{\beta}_c _{\text{med}}$ (IQR)	β_c p95	$f_{10^\circ}^{\text{origin}}$
S08-robust	22° (40)	26° (10.5)	44°	0.00
S09-primary	16° (17.5)	24° (9)	48°	0.00

Table S5

Test 2a results. $f_{10^\circ}^{\text{origin}} = 0/140$ for both candidates: no sample was recovered within 10° of the origin under a null ground truth. Effective parameter uncertainty (PCA-basis): $\sim 20^\circ$ (β_s) and $\sim 25^\circ$ (β_c).

Candidate	rank _{dist}	Verdict
S08-robust	0.875	FAIL
S09-primary	0.875	FAIL

Table S6

Test 2b results. Neither candidate exceeds all HC pseudo-CVD distances.

Candidate	Real loss	Perm 5% cut	p_{perm}
S08-robust	-2.892	-3.136	0.167
S09-primary	-1.681	-3.053	0.471

Table S7

Test 2c results. Neither candidate reaches significance against the label-permutation null.

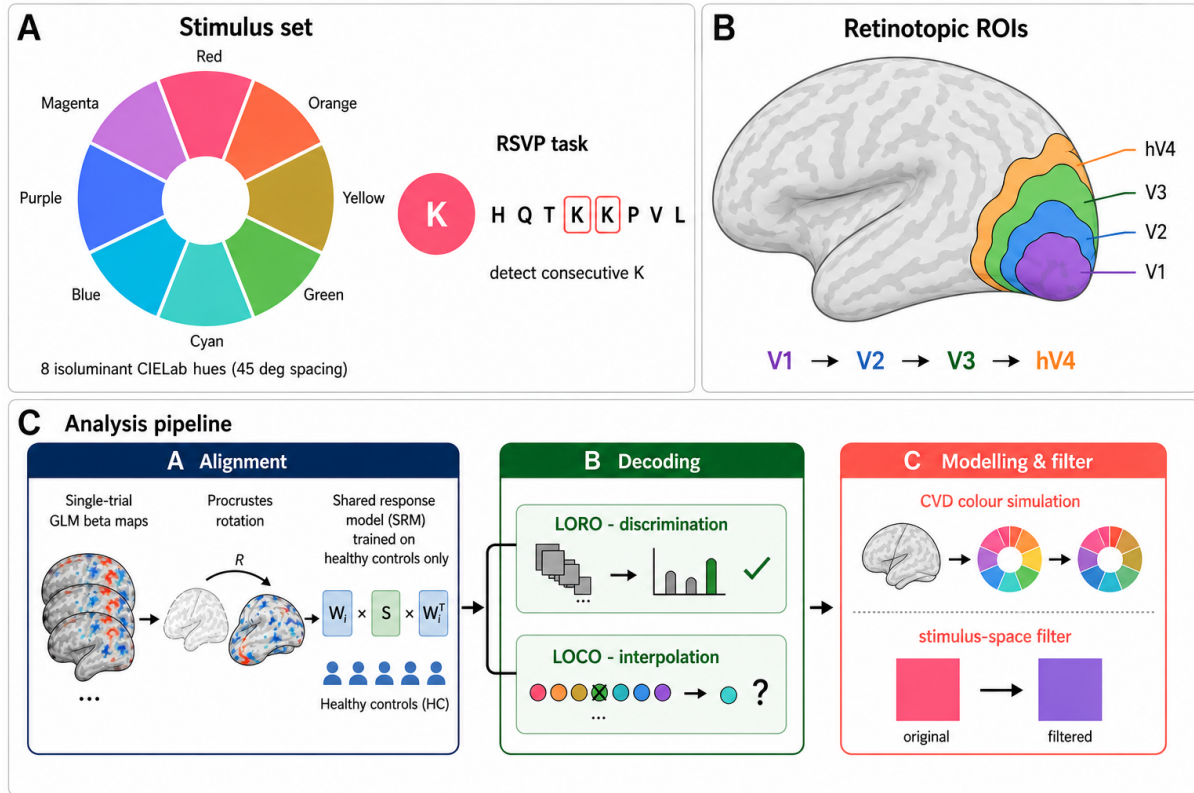


Figure 1. Experimental paradigm and analysis pipeline. (A) Stimulus set: 8 isoluminant CIE $L^*a^*b^*$ hues at 45° angular spacing ($L^* = 75$, chroma = 40; red–magenta). (B) Retinotopic ROIs: V1, V2, V3, hV4 (standard retinotopic mapping). (C) Analysis pipeline. Stage A: two-stage GLM single-trial β amplitudes, Procrustes-aligned across runs, projected into an SRM space trained on HC participants only. Decoding: LORO (discrimination), LOCO (interpolation). CVD characterization: 2-component cortical model with S-cone-axis (β_s) and confusion-axis (β_c) components. Pre-image yields per-subject stimulus-space filter. HC: $n = 7$ (sub-01–07); deutan: the deutan participant; protan: the protan participant.

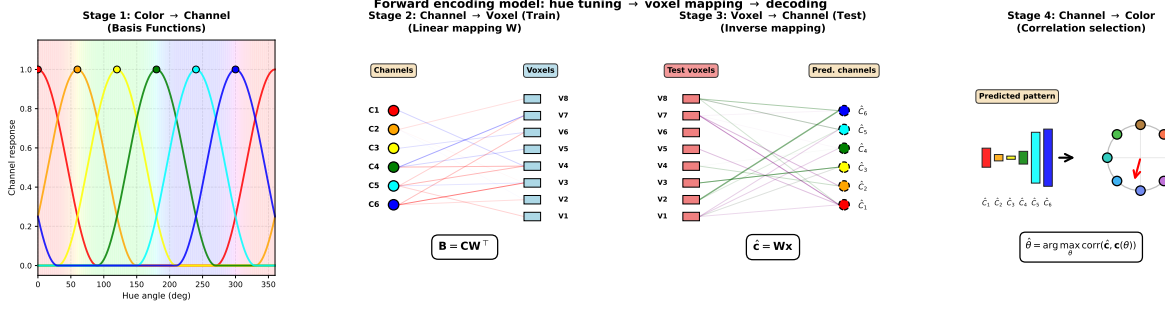
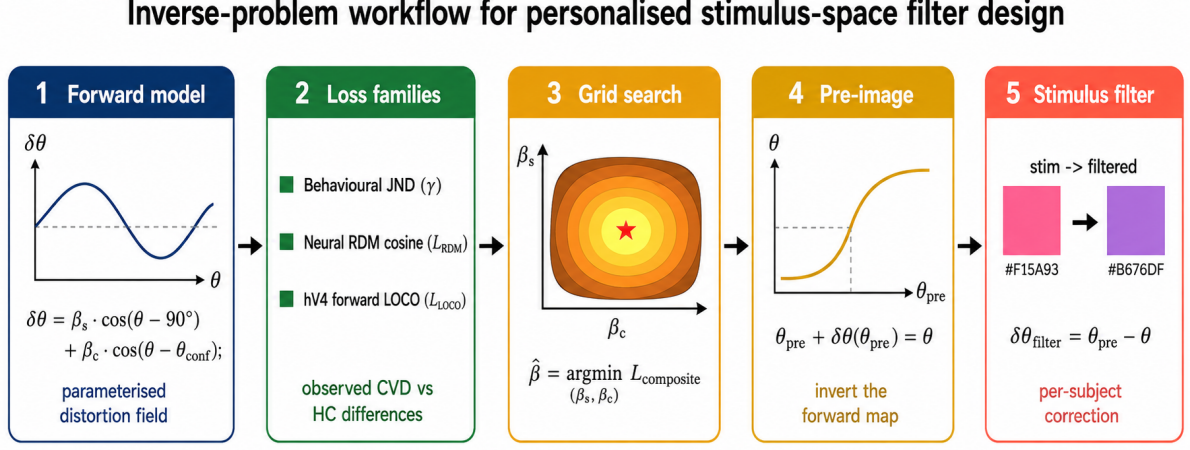


Figure 2. Forward encoding model schematic. Stage 1: each displayed hue is encoded as channel activations by six half-wave rectified squared-cosine basis functions, with preferred hues spaced at 60 degrees (Brouwer & Heeger 2009). Stage 2 (training): the channel-to-voxel weights W are fit by ordinary least squares for hue decoding, or by GCV-regularized ridge regression for voxel-response prediction (encoding). Stage 3 (testing): for a held-out voxel pattern, channel activations are reconstructed as $\hat{\mathbf{c}} = W\mathbf{x}$. Stage 4: the decoded hue is recovered as the angle whose basis vector has the highest Pearson correlation with the reconstructed channel pattern. The same model serves both decoding schemes (Section 4.7): cross-run discrimination (LORO) and cross-color interpolation (LOCO). Their basis and correlation-based selection are identical, and only the held-out test set differs. Decoding uses the OLS pseudoinverse weights.



Schematic; glyphs illustrative

Figure 3. Inverse-problem workflow for personalized stimulus-space filter

design. (1) A two-parameter forward model expresses the per-hue distortion as $\delta\theta(\theta) = \beta_s \cos(\theta - 90^\circ) + \beta_c \cos(\theta - \theta_{\text{conf}})$: an S-cone-axis component (β_s) plus a family-specific confusion-axis rotation (β_c ; $\theta_{\text{conf}} = 16^\circ$ protan, 150° deutan). (2) Three loss families quantify CVD–HC differences — behavioural JND ratios (γ), neural representational ΔRDM cosine, and hv4 forward-encoding LOCO; only the LOCO loss couples to the forward encoder. (3) A grid search over (β_s, β_c) minimizes the z -scored composite to the selected fit $\hat{\beta}$. (4) The fitted forward map is analytically inverted: for each displayed hue θ , the corrected stimulus θ_{pre} solves $\theta_{\text{pre}} + \delta\theta(\theta_{\text{pre}}) = \theta$. (5) The per-subject filter is $\delta\theta_{\text{filter}}(\theta) = \theta_{\text{pre}} - \theta$. Glyphs are illustrative; the fitted loss landscape is shown in Figure 6.

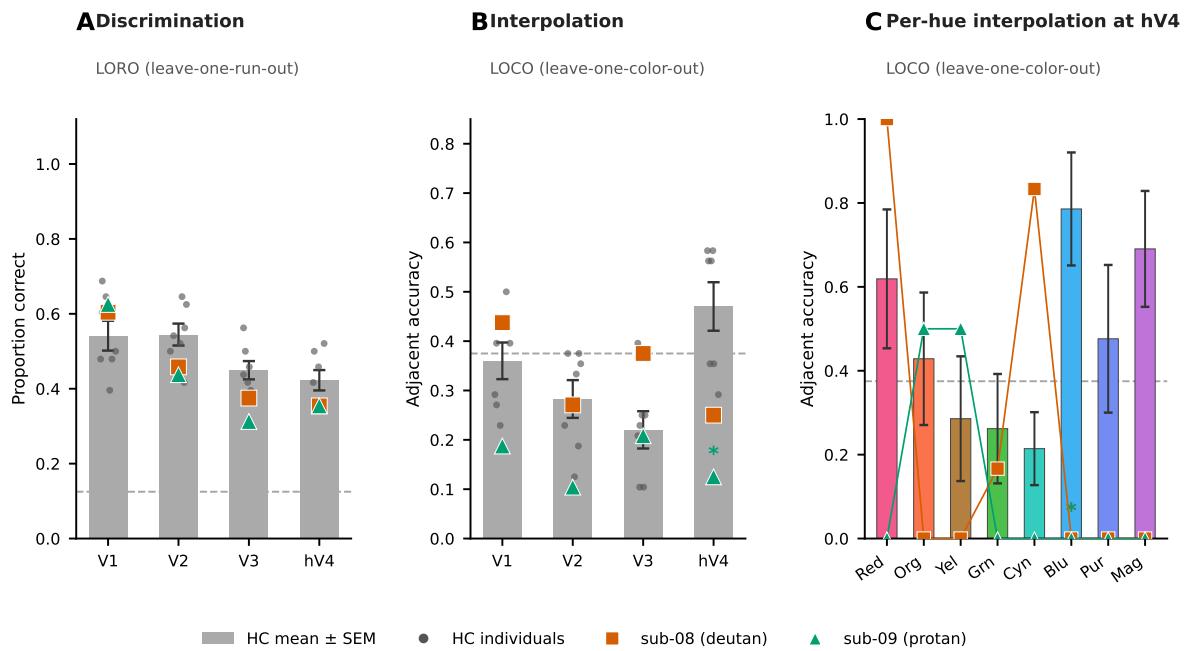


Figure 4. Color discrimination is preserved but interpolation is reduced at hV4 in CVD. (A) LORO discrimination accuracy (forward encoding model, 8-class, SRM-aligned), V1–hV4. HC mean $\pm$ SEM ($n = 7$; $n = 6$ at hV4, sub-07 excluded for low voxel count; gray); the deutan participant (orange square), the protan participant (teal triangle). Chance = $1/8$. *(B)* LOCO adjacent accuracy (± 1 hue step; chance = $3/8$). *: the protan participant below HC at hV4 (Crawford & Howell $t = -2.91$, $p = 0.017$; exploratory single-case finding); the deutan participant $t = -1.84$, $p = 0.063$, n.s. *(C)* Per-hue adjacent accuracy at hV4, HC bars $\pm$ SEM ($n = 6$). Per-hue single-case tests (Crawford & Howell, one-tailed, uncorrected and exploratory) reached significance at no individual hue; the deficit concentrated on the S-cone intermediate hues blue, purple, and magenta, where both participants showed zero adjacent accuracy. Joint pattern in (A) and (B–C) defines the filter precondition and the functional vulnerability profile, respectively.

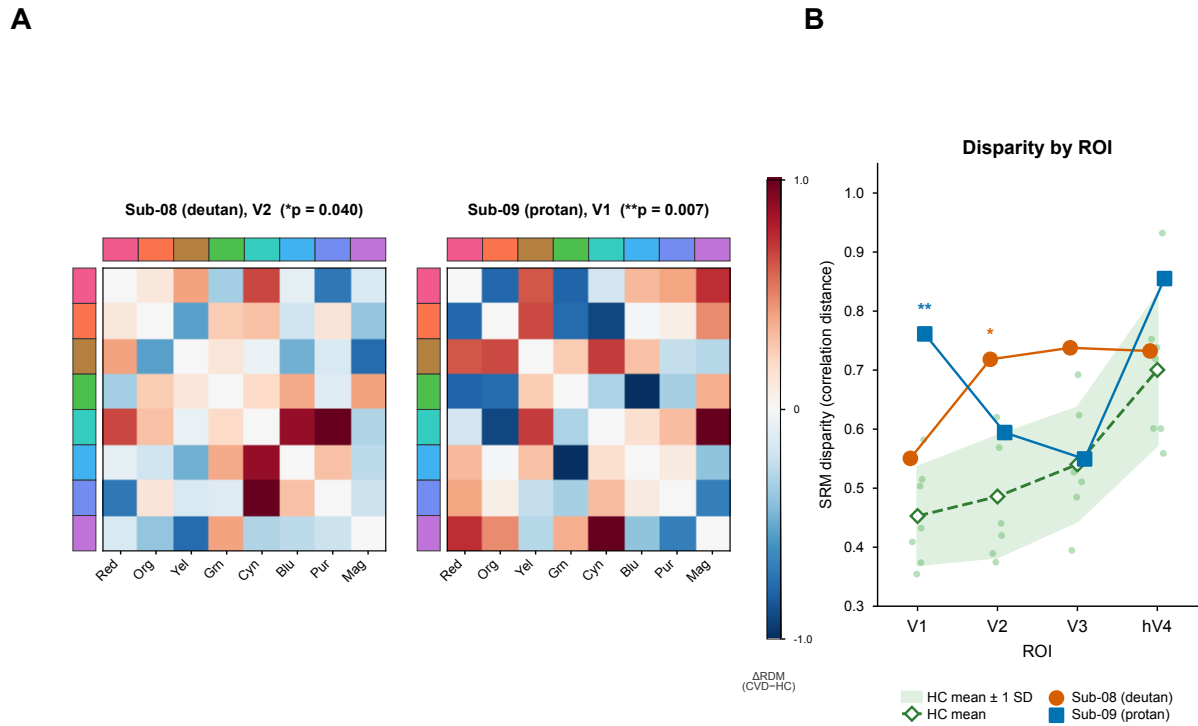


Figure 5. Each CVD subject shows elevated color-representation disparity at a distinct ROI. (A) $\Delta RDM = RDM_{CVD} - \overline{RDM}_{HC}$ in SRM-aligned space at each subject's primary ROI (the deutan participant V2, the protan participant V1). Warm: CVD distances larger than HC mean; cool: smaller. **(B)** Mean pairwise correlation distance (disparity) per subject per ROI. HC band: group mean $\pm$ 1 SD ($n = 7$); individual HC LOO as dots. Crawford & Howell tests in the common HC space (the deutan participant V2: $p = 0.040$; the protan participant V1: $p = 0.007$); under a symmetric LOSO control the protan participant V1 remains significant ($p = 0.045$) and the deutan participant V2 attenuates to a trend ($p = 0.116$). Exploratory single-case. (A) shows the directional pattern of pairwise deviations. (B) tests the absolute amplitude of pairwise distances against the HC LOO distribution. Each subject showed elevated disparity at a single, distinct ROI (the deutan participant V2; the protan participant V1).

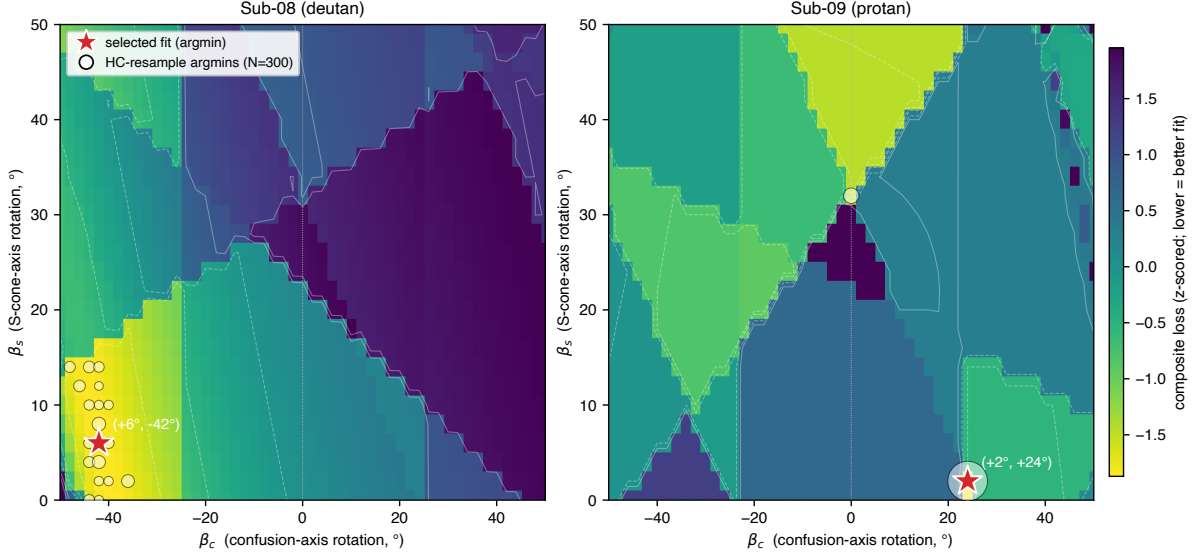


Figure 6. Per-subject loss landscape and parameter uncertainty. z -scored composite loss over the (β_s, β_c) grid for each CVD participant’s selected loss combination (the deutan participant: $\gamma_{\text{OY}} + L_{\text{RDM}}^{(V2)}$; the protan participant: $\gamma_{\text{all}} + L_{\text{RDM}}^{(V1)}$), reconstructed on the full 7-HC pool. Lower (yellow) indicates better fit. Red star: selected fit (argmin) — the deutan participant $(\hat{\beta}_s, \hat{\beta}_c) = (6^\circ, -42^\circ)$; the protan participant $(2^\circ, +24^\circ)$. White points: per-resample argmins over $N = 300$ HC 5-train/2-test resamples, showing parameter uncertainty. The selected fits are descriptive low-dimensional embeddings of each participant’s distortion, not physiological point estimates (this section).

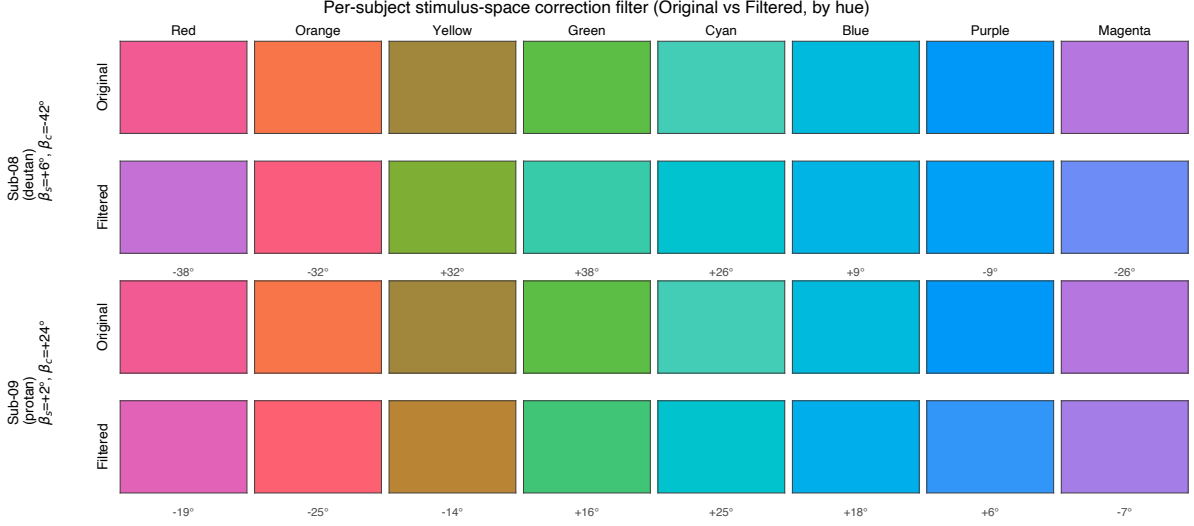


Figure 7. Per-subject stimulus-space correction filter. For each CVD participant, the displayed stimulus (*Original*) and the corrected stimulus (*Filtered*) are shown for the eight scanner hues. Left block: the deutan participant, $(\hat{\beta}_s, \hat{\beta}_c) = (6^\circ, -42^\circ)$. Right block: the protan participant, $(\hat{\beta}_s, \hat{\beta}_c) = (2^\circ, +24^\circ)$. The filtered stimulus is the exact numerical pre-image of the fitted 2-component transform; the per-hue correction $\delta\theta_{\text{filter}} = \theta_{\text{pre}} - \theta$ is annotated beneath each *Filtered* swatch (8/8 pre-images exact, residual $< 0.001^\circ$ for both subjects). Rendering coordinate space: STIM_LAB CIELab (project convention). Neural validation (hue-interpolation adjacent accuracy, forward-tuning, and representational geometry) and behavioural validation (JND thresholds and 8AFC color identification) in a second deutan session are reported in fig. 8; only a separate two-alternative forced-choice (2AFC) fine-discrimination test remains for future work.

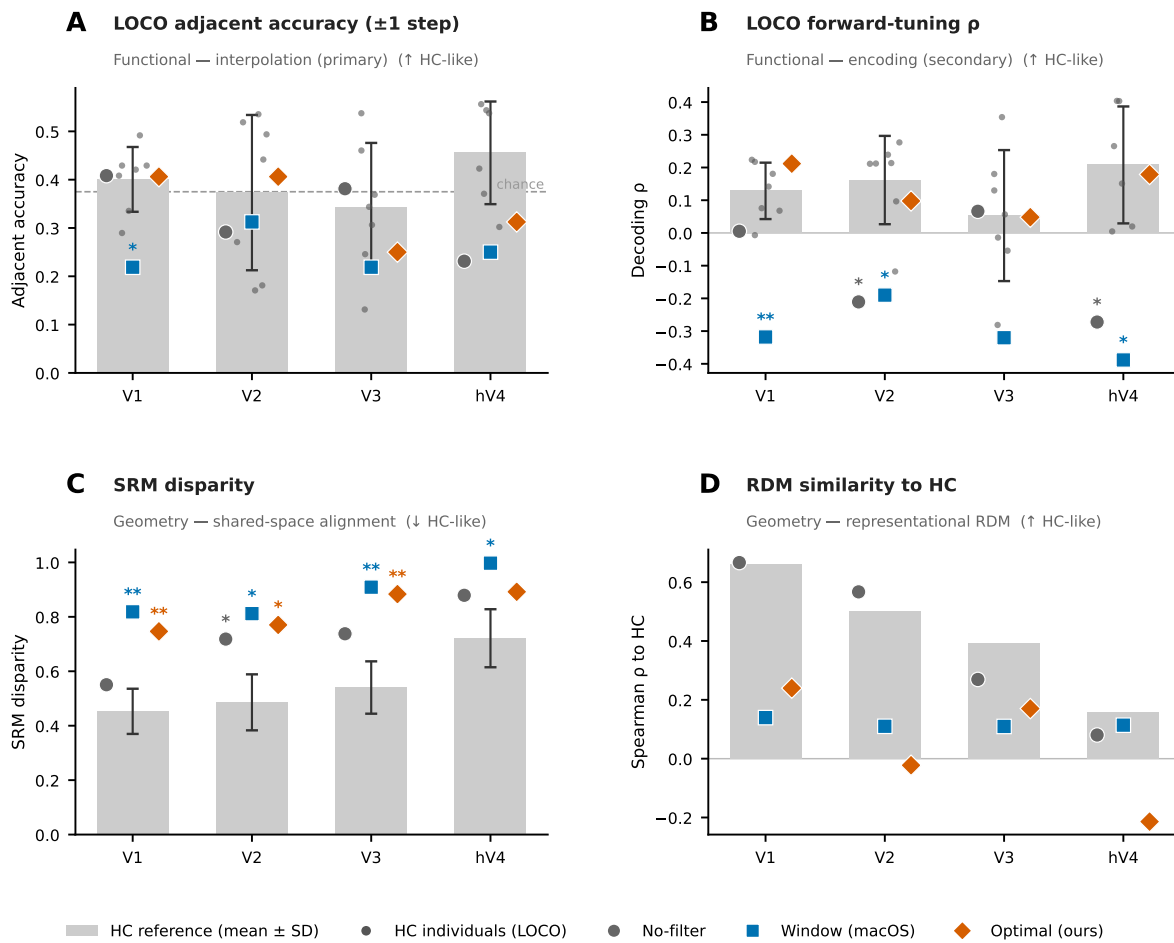


Figure 8. The personalized filter partially restores hue interpolation toward HC but not an HC-like color geometry (the deutan participant, second session). Functional indices (A, B) and representational geometry (C, D) across V1–hV4. Gray bars, HC reference (mean $\pm$ SD, $n = 7$); dots in (A), individual HC values. Markers: no-filter (gray circle, Session-1 baseline), deployed macOS filter (blue square), personalized filter (orange diamond). **(A)** LOCO adjacent accuracy (± 1 hue step; the Session-1 primary interpolation metric; $\uparrow$ HC-like). hV4 is the validated ROI; chance = $3/8$. **(B)** LOCO forward-tuning correlation to the canonical hue map (secondary encoding index; $\uparrow$ HC-like). **(C)** SRM disparity into the HC shared space ($\downarrow$ HC-like). **(D)** Shared-space correlation-distance RDM similarity to HC ($\uparrow$ HC-like). Single subject, descriptive. Stars, Crawford–Howell single-case deviation from the HC distribution ($*p < .05$, $**p < .01$, $***p < .001$).

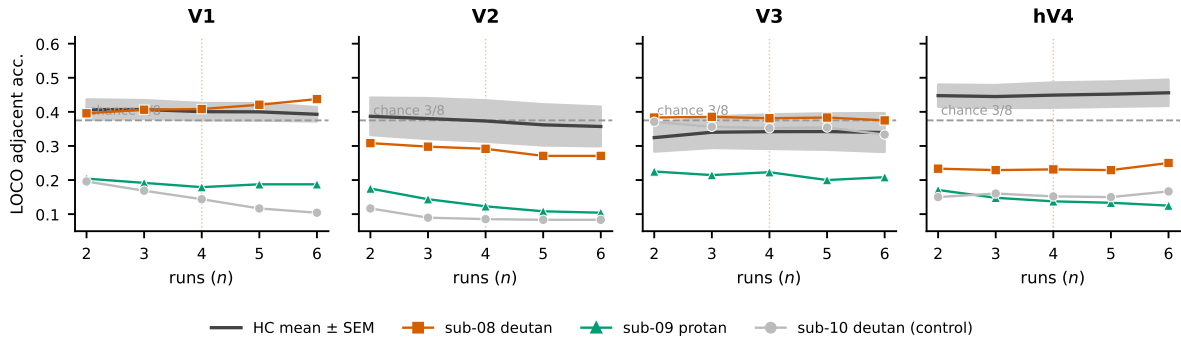


Figure 9. The primary interpolation metric is stable under run-count reduction. LOCO adjacent accuracy (canonical FE-6 basis, OLS pseudoinverse decoder) as a function of the number of runs n , computed over all $\binom{6}{n}$ run subsets of the Session-1 data, per ROI. Black line, healthy-control mean $\pm$ SEM (shaded band); dashed line, the 3/8 chance level; markers, the two CVD single cases (sub-08 deutan, sub-09 protan) and the deutan control sub-10. At hV4 the healthy-control mean stays above chance and the CVD deficit persists at every run count, including the deployed $n = 4$ (orange guide); the metric does not depend on having the full six runs.